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PERFORMANCE SPECIFICATION

HEATER, ARMY SPACE (ASH), ELECTRIC POWERED, MULTIFUEL

This specification is approved for use within the USA Communications-Electronics Command, Logistics and Readiness Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Army Space Heater (ASH), Electric Powered, Multifuel, hereinafter referred to as "ASH", used to provide adequate temperatures in mobile shelters.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are needed to meet the requirements specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Communications-Electronics Command, CECOM RDE Center, ATTN: AMSEL-LC-IEW-D-ED, 10115 Gridley Rd, Ste 228, Ft Belvoir VA 22060-5849 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4520

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SPECIFICATIONS

FEDERAL

A-A-50271	-	Plate, Identification.
A-A-52557	-	Fuel, Diesel; For Posts, Camps and Stations.

DEPARTMENT OF DEFENSE

MIL-T-83133	-	Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-461	-	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of.
MS3401	-	Connectors, Receptacle, Electric, Wall Mounting, Front Release, Crimp Contact, AN Type.
MS3406	-	Connector, Plug, Electric, Front Release, Crimp Contact, AN Type.

(Unless otherwise indicated, copies of above specifications and standards are available from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents and publications. The following other Government documents and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATIONS

SEL Form 1183	-	System Safety Design Verification Checklist.

(Copies of the above publication are available from: Commander, USA CECOM Safety Office, ATTN: AMSEL-SF-SE, Ft Monmouth, NJ 07703-5024.)

SUPPLY CATALOG

SC 5180-90-CL-N26	-	Tool Supply Catalog, Mechanics Kit.

(Copies of other publications and drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Communications-Electronics Command, CECOM RDE Center, ATTN: AMSEL-LC-IEW-D-EE, 10115 Gridley Rd Ste 228, Ft Belvoir VA 22060-5849.)

CODE OF FEDERAL REGULATIONS

29 CFR-1910	-	Occupational Health - Permissible Exposure Limits.

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the

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issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS 1369	-	Duct, Air, Flexible and Semi-Rigid.
NAS 1370	-	Duct, Air, Flexible, Helix Wire Supported.

(Application for copies should be sent to the Aerospace Industries Association of America, 1250 Eye St., NW, Washington, DC 20005.)

AMERICAN CONFERENCE OF GOVERNMENTAL INDUSTRIAL HYGIENISTS (ACGIH)

Threshold Limit Values and Biological Exposure Indices.

(Application for copies should be addressed to American Conference of Governmental Industrial Hygienists, 6500 Glenway Avenue, Building D-7, Cincinnati, OH 45211-4438.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ASQC Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes.
S1.4	-	Meters, Sound Level, Specification For.
S1.11	-	Octave-Band and Fractional-Octave-Band Analog and Digital, Filers, specification For.
S1.40	-	Calibrators, Acoustical, Specification For.
Z535.4	-	Product Safety Signs and Labels.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR CONDITIONING ENGINEERS, INC. (ASHRAE)

Standard 45	-	Methods of Testing for Rating Non-Residential Warm Air Heaters.

(Application for copies should be addressed to the ASHRAE Publications Sales Department, 1791 Tullie Circle, NE, Atlanta, GA 30329.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Qualification Standards for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators.

(Application for copies should be addressed to: American Society of Mechanical Engineers, 345 E 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 2276 - Particulate Contaminate in Aviation Fuel by Line Sampling.

(Application for copies should be addressed to: American Society for Test and Materials, 1916 Race Street, Philadelphia PA 19103.)

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AMERICAN WELDING SOCIETY, INC. (AWS)

B2.1 - Standard for Welding Procedure and Performance Qualification.
B2.2 - Standard for Brazing Procedure and Performance Qualification.
D1.1 - Structural Welding Code - Steel.
D1.2 - Structural Welding Code - Aluminum.

(Application for copies should be addressed to the American Welding Society, Inc., 550 LeJeune Road, PO Box 351040, Miami, FL 33135.)

ASSOCIATION OF AMERICAN RAILROADS (AAR)

Section 6 - Rules Governing the Loading of Department of Defense Materiel on the Open Top Cars.

(Application for copies should be addressed to: Association of American Railroads, 50 F Street NW, Washington, DC 20001.)

COMPRESSED GAS ASSOCIATION, INC. (CGA)

Pamphlet G-7.1 - Commodity Specification for Air.

(Application for copies should be addressed to the Compressed Gas Association, Inc. 1235 Jefferson Davis Highway, Suite 501, Arlington, VA 22202.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electric Code.

(Application for copies should be addressed to the National Fire Protection Association, Battery March Park, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J184 - Systems, Acquisition, Qualifying a Sound Data.
J429 - Mechanical and Material Requirements for Externally Threaded Fasteners.
J492 - Rivets and Riveting.
J1362 - Graphical Symbols for Operator Controls and Displays on Off-Road, Self-Propelled Work Machines.

(Application for copies should be addressed to: Society of Automotive Engineers, Inc, 400 Commonwealth Drive, Warrendale PA 15096.)

UNDERWRITERS LABORATORIES, INC. (UL)

969 - Standard for Marking and Labeling Systems.
1950 - Equipment, Safety of Information Technology, Including Electrical Business Equipment.

(Application for copies should be addressed to: Underwriters Laboratories Inc., 333 Pfingsten Rd, Northbrook IL 60062.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The ASH heater shall be electric powered; re-circulating; operate with either diesel or JP-8 fuels; controlled by a remote thermostat; and require no greater than 3 kilowatts (kW) of external power. The heater shall deliver clean, heated and ventilated air through sealed detachable flexible ducts; have a protective enclosure; have wheels and handrails or handles for local movement and positioning; and withstand environmental conditions of arctic use.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.3 Performance.

3.3.1 Rated output and efficiency. The heater shall deliver not less than 120,000 British Thermal Units per Hour (BTUH) of clean air (see 3.8.7) when operated in the all fresh air mode, with an air temperature of 0 °F, at sea level, against an 0.5-inch water gauge (WG) pressure, using diesel fuel, grade Low Sulfur No. 1-D conforming to A-A-52557. For this rated output, the temperature at the output duct connection on the heater shall be between 120 and 180 °F. The overall heater efficiency (output/input) shall not be less than 80 percent. The indicated efficiency of the heater shall not be less than 85 percent.

3.3.2 Dimensions. The overall dimensions of the complete heater assembly with accessories stowed for packing shall not exceed the following:

Height: 58 inches
Length: 65 inches
Width: 40 inches

Separate duct storage is acceptable and shall not be included as part of the overall dimensions.

3.3.3 Weight. The heater shall weigh no more than 355 pounds, including the weight of accessories, lubricants, and internal fuel sufficient for one hour of operation. The weight of the ducts shall not be included in the weight specified.

3.4 Environmental.

3.4.1 Storage and transportation conditions. The heater shall withstand storage and transportation without impairment of capabilities under climatic conditions ranging from -60 to +160 °F and to altitudes of 4,000 feet.

3.4.2 Operating conditions. The heater shall start and operate in all modes ("VENT", "AUTO", "MANUAL") within an ambient air temperature ranging from -40 to +65 °F.

3.4.3 Humidity and salt fog. When subjected to the tests specified in 4.6.4.3 and 4.6.4.4, the heater shall show no evidence of peeling paint, malfunctioning, rough operation, electrical shorts, erratic pressures, or any other irregular operation.

3.4.4 Rain. The heater shall comply with the following:

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a. The heater shall start and operate during the last ten minutes of the rain test.
b. There shall be no penetration of water into/onto any electrical component or connection whatsoever.
c. There shall be no water absorption by the duct assembly or other damage that might affect subsequent operation of the duct assembly.

3.4.5 Fungus and moisture resistance. The electrical circuitry, including all components and connections, shall be protected from the effects of fungus growth and moisture. Hermetically sealed components need not be treated for fungus growth and moisture.

3.4.6 Noise limits. The noise level at the operator's position, defined to be 18 inches from the control panel, and other personnel occupied areas of the heater (defined to be anywhere within 1 meter from the perimeter of the heater at locations other than the operator's position) shall not exceed 85 db(A). The noise level at 18 inches from each duct outlet shall not exceed 65 db(A).

3.4.7 NBC compatibility. The heater design shall be compatible with NBC protection and shall permit performance of mission-essential operations, maintenance, and decontamination tasks by suitably clothed, trained and acclimatized personnel for the survival periods and NBC environments required by the system.

3.5 Electric power. The heater shall start and operate when provided with nominal 120 volt, single phase, 50/60 Hz power, in all modes, with the following power limitations:

a. Inrush - The inrush current draw of the heater during start-up shall be less than or equal to 45 amperes (amps). The 45 amps limit shall be maintained until all starting devices are non-operating and the heater is operating under rated conditions (steady-state).

b. Steady-state - The current draw of the heater shall not exceed 24 amps, with voltage maintained at 120 volts.

3.5.1 Electromagnetic interference (EMI). Radiated emissions, electric field, 10 kHz to 18 GHz, (RE102) electric field emissions shall not be radiated from the heater or any interconnecting cables in excess of those shown in figure 1. Above 30 MHz, the limits shall be met for both horizontally and vertically polarized fields.

3.5.2 Power cable. An external 25-foot power cable shall be supplied with the heater. The cable shall be rated for outdoor use and shall meet all the environmental and electrical requirements specified herein. One end of the cable shall be attached to the heater; the other end of the cable shall have a power connector conforming to MS3406, part number MS3406D16-10P. A captive chained or cabled cover for the connector and a strain relief cable clamp with weatherproof sleeve shall be provided.

3.5.3 Lead assembly. A lead assembly shall be supplied with the heater and shall conform to figure 2. The lead assembly shall include a strain relief, cable clamp, and a cover for the electrical connector. The cover shall be captive chained or cabled to the connector to prevent loss of the cover.

3.6 Fuel system. The fuel system shall permit the heater to operate continuously without flooding, with the fuel container 12 inches below the

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base of the heater and up to 10 feet above the heater, and shall include the following:

a. An integral fuel tank of sufficient capacity for operation of the heater at maximum rated heat output for a minimum of 8 hours. The fuel tank shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments. The fuel tank shall be equipped with an overflow vent, water drain, quantity gauge, a removable filler strainer, and a filler neck/cap recessed flush with the enclosure exterior surface. The cap shall be captive chained or cabled to prevent loss of the cap. The chain or cable shall permit unobstructive refueling of the tank. Fuel tanks shall be provided with a drain connection separated from the fuel supply line. The drain connection shall be leakproof. The fuel tank shall be compatible with the fuels specified in 3.6.1. During transportation and storage the interior of fuel tanks shall be preserved by one of the following methods as applicable.

(1) The interior of the steel fuel tanks shall be injected with a rust-preventive fog oil. Filler neck screen and cap shall be coated with the same preservative oil. After the oil-fogging treatment, drain any excess of the preservative oil from the fuel tank. Replace fuel tank drain plugs and properly tighten.

(2) Unless otherwise specified, the interior of certain types of fuel tanks (aluminum, fiberglass, and tanks with synthetic rubber liner) require no preservation other than draining the fuel.

b. A manual valve to permit operation from the external fuel supply or the onboard operating tank.

c. A 5/16-inch inner diameter 25-foot external fuel hose, conforming to SAE standards for low pressure for fuel transfer from an external fuel supply. The hose shall have captive protective caps. One end of the hose shall have quick disconnect fittings for connection to the heater. For connection to the external fuel supply, the other end of the hose shall be provided with a fitting compatible for mating with a nipple, flared tube and pipe thread, 5/16 tube OD X 1/4 ANPT X 1/2-20 UNF-3A.

d. A quick disconnect fitting, flush mounted on the heater exterior, for mating with the external fuel hose.

3.6.1 Multifuel capability. The heater shall meet requirements specified herein using diesel fuels conforming to A-A-52557 and turbine fuel conforming to MIL-T-83133 within the following temperature ranges:

Ambient Temperature	Specification	Grade
Above +20 °F	A-A-52557	Low Sulfur No. 2-D
Above -25 °F	A-A-52557	Low Sulfur No. 1-D
Above -50 °F	MIL-T-83133	JP-8

3.6.1.1 Ozone-depleting chemical (ODC) note. The specification for JP-8, MIL-T-83133, calls out a test method for particulates (ASTM D2276). This method requires the use of an ozone-depleting chemical (ODC), trichlorotrifluoroethane (CFC-113), for cleaning of the test apparatus. This chemical shall not be used. Several non-ODC substitutes are commercially available (see note 6.6).

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3.7 Transportability. The heaters shall be capable of being transported by military and commercial trailers, trains, marine vessels, and aircraft; and must be capable of withstanding the conditions encountered in shipment without damage or permanent deformation (see 3.7.2, 3.7.3, and 3.7.4). The heaters shall be equipped with tiedown and slinging provisions.

3.7.1 Slinging and tiedown provisions. The heater shall have four slinging provisions and no less than four tiedown provisions. A combination of multipurpose, slinging and tiedown, provisions may be used to meet this requirement. Slinging provisions shall be located above the heater's center of gravity (CG). Wheel hubs shall not be designed for or used as slinging or tiedown points. Provisions that can be removed without the use of tools shall not be used. All slinging and tiedown provisions shall be designed to prevent movable parts from freezing in place during cold weather. The lift provisions shall be non-protruding and self concealing. The minimum inside dimension of each provision shall be 3.0 inches. Provisions shall be designed to swivel or rotate and the minimum inside dimension shall be met throughout the full range of motion. Slinging provisions shall enable the complete heaters to be lifted in the normal operating position. The provisions shall be located so that not less than 1-inch clearance is maintained between slings and all exterior parts and shall be fastened to members which will withstand stresses in the amount and direction of pull specified for the provisions. All slinging/tiedown provisions shall be labeled "LIFT", "TIEDOWN", or "LIFT/TIEDOWN", as applicable, in 1-inch (2.54 cm) high letters.

3.7.2 Provisions strength. Slinging provisions shall be capable of withstanding 2.3 times the static load induced by the heater alone without failure (see 6.3.2), deformation or damage to any of the lift provisions, or any structural damage to the heater. The tiedown provisions shall withstand the following loading: 4.0 times the gross weight of the heater in the forward and aft direction of the longitudinal axis of the heater; 2.0 times the gross weight in the downward direction of the vertical axis of the heater; and 1.5 times the gross weight in each direction of the lateral axis of the heater. The tiedown provisions shall satisfactorily complete the pull testing as specified in 4.6.5. If multipurpose provisions are used, they shall meet all of the strength requirements herein.

3.7.3 Vibration. During operation and testing, the heater shall show no evidence of weld failure, permanent deformation, cracking, loosening or breaking of the provision or its connecting structural components, visual misalignment or loosening of components, malfunction of components or controls, or fuel leaks.

3.7.4 Rail impact. The heaters shall be capable of withstanding shock loads resulting from rail impact testing in accordance with 4.6.13 without failure, damage, or permanent deformation.

3.7.5 Mobility/parking devices. The heater shall be mounted on a chassis with integral skids and pneumatic tires, and the tires shall be mounted on wheels that are removable or retractable without the use of external tools. Provisions shall be made for securing the wheels to the heater when not installed for use. The heater shall have continuous handrails or minimum handles to assist in manual localized movement and positioning. The heater shall be provided with a parking stand to provide three-point support when the wheels are installed and with integral means to prevent movement of the heater when parked in level or specified inclined positions (see 3.7.6). Weight distribution of heater components shall provide a safe and stable configuration in all specified operating positions.

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3.7.6 Inclined operation. The heater shall start and operate while inclined on any axis and at any angle up to 10 degrees without fuel leakage or damage to components.

3.8 Combustion system. The combustion system shall provide for the functions of the burner, combustion chamber, heat exchanger, and exhaust stack. The combustion system shall be designed for continuous, safe and efficient combustion of the fuels specified in 3.6.1 for all specified conditions of heat output, ambient temperature and barometric pressure. The combustion system shall not fail from exposure to any condition encountered during operation at any condition specified herein, or during testing specified herein. These conditions shall include, but not be limited to, extreme operating temperatures, corrosion, and rough handling and transport. The system shall be designed to permit disassembly and cleaning.

3.8.1 Burner. The burner shall provide a means of introducing and mixing fuel and combustion air for ignition and initiation of the combustion process.

3.8.2 Emissions. When operating under the following conditions, the burner shall not produce smoke from the combustion exhaust in excess of Bachrach no. 3: full load, steady-state, at altitudes up to 4,000 feet, temperature ranging from -40 to +65 °F and burning each of the fuels listed in 3.6.1.

3.8.3 Combustion chamber. The combustion chamber shall be a separable, removable chamber or a part of the heat exchanger and shall serve to continue mixing and completion of the combustion process. An observation port shall be provided for the purpose of observing the ignition point and the combustion process. The observation port shall not leak combustion gases to the heated air, and shall be so located that it does not become clogged or clouded by the products of combustion.

3.8.4 Flame failure. The flame failure control shall respond directly to flame properties, sensing the presence or absence of flame after the ignition cycle. Flame failure after the ignition cycle shall cause the control to shut down the burner within 15-seconds and illuminate the panel indicator light. After shutdown, the control shall lock out the burner and require reset from the control panel. Blower fan(s) for air circulation shall continue operation in the event of a flame failure (see 3.9.3).

3.8.5 High temperature limit. A high temperature limit control shall shut down the burner when the temperature at the heater discharge outlet exceeds the pre-set limit and illuminate the panel indicator light. After a shutdown, the control shall reset automatically after the outlet temperature falls to 10 ±5 °F below the shutdown temperature. The pre-set limit on the control shall be 200, 10 +0 °F.

3.8.6 Heat exchanger. The heat exchanger shall provide for the necessary heat transfer between the combustion gases and the ventilating air to be heated. The heat exchanger shall prevent combustion products from entering the ventilating airstream (see 3.8.7). The heat exchanger shall be designed for maximum efficiency consistent with maintaining the discharge temperature of the exhaust gases above the dewpoint in the exhaust stack. The heat exchanger shall be capable of draining condensate overboard. No condensate shall be allowed to collect on the heater surface.

3.8.7 Heated air quality. Combustion by-products in the heated air shall be within the specification requirement of grade D, type I compressed air in accordance with CGA Pamphlet G-7.1; the permissible exposure limits specified

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by 29 CFR 1910; and the limits specified in the ACGIH's Threshold Limit Values. Intakes for ventilation systems shall be located to minimize the introduction of contaminated air from exhaust stack.

3.8.8 Exhaust stack. A detachable, vertical exhaust stack extension, not less than 18 inches long, shall be provided and located to minimize exhaust discharge from entering the fresh air intake. The extension shall have an outer protective surface to prevent burns from accidental contact. Provision shall be made for secure stowage of the exhaust stack extension on the heater for packing and movement.

3.9 Controls. Controls, switches and indicators shall provide for safe and efficient operation of the heater and shall be located on a central control panel, readily visible and accessible from the top, during operation of the heater, by opening a small cover. An indication of control activation shall be provided (e.g., snap feel, audible click, or indicator light illumination). All switches and controls shall be labeled for identification and settings (see 3.16 and 3.17.1) and shall be operable with arctic mittens (see 3.17.1).

3.9.1 Start-stop. A switch shall be provided to start and stop heater operation. A green illumination light shall be provided to indicate when power is applied to the heater.

3.9.2 Operating mode control. A switch shall be provided for selecting and setting each of the following operating modes:

a. "VENT" Mode - continuous circulation of fresh and/or re-circulated unheated air.
b. "AUTO" HEAT Mode - continuous air circulation with burner and air temperature controlled by the remote thermostat.
c. "MANUAL" HEAT Mode - continuous air circulation with the burner controlled by the switch setting and safety controls (see 3.8.4 and 3.8.5).

3.9.3 Flame failure light and reset. A red indicator light located on the control panel shall illuminate upon occurrence of a flame failure. A reset button shall be provided on the control panel to reset the flame failure control (see 3.8.4). The indicator shall be labeled "FLAME OUT". The reset button shall be labeled "FLAME RESET". The indicator and reset button shall be in close proximity and readily accessible on the heater.

3.9.4 High temperature light. A red indicator light located on the control panel shall illuminate upon occurrence of an over-temperature condition and extinguish when the control resets (see 3.8.5). The indicator light shall be labeled "HIGH TEMP".

3.9.5 Remote thermostat. A hermetically sealed remote thermostat, with a manual override, and minimum range from +35 to +85 °F shall be provided. The thermostat shall have a maximum differential of 5 °F. The thermostat shall be assembled with a cable of 25 feet minimum length and an environmental resistant connector compatible with a mating receptacle on the heater.

3.9.6 Hourmeter. An hourmeter shall be installed so that it is visible at the control panel. This meter shall record burner-on time and shall perform as follows:

a. The life accuracy shall be ±0.1 percent from 102 to 128 V at 60 Hz and -65 to +212 °F.

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b. Meter shall operate satisfactorily and maintain the life accuracy limits specified above over the frequency range of 50 to 70 Hz (Note: Accuracy is degraded at the same percent as the percentage change in input frequency from 60 Hz, plus the specified accuracy tolerance.)
c. Meter flange (if a separate part from the enclosure) shall be fastened to the enclosure.
d. Power consumption at 155 V, 60 Hz AC shall not exceed 4.0 watts.
e. Operating temperature range shall be -65 to +212 °F
f. Total time range readout of the meters shall be 9,999 hours.
g. Operating voltage range shall be from 102 to 128 V AC.

3.9.7 Human factors engineering of controls. Controls shall be located within the operator's functional reach and view without the need to assume an uncomfortable or unusual posture. Control design shall facilitate accurate and efficient system operation.

3.9.7.1 Consistency of movement. Direction of control movement shall be consistent with the effect of change. For example, the movement of a control forward, clockwise, to the right, up or pressing a control, shall turn the equipment or component on, cause the quantity to increase, or cause the effected element to move forward, clockwise, to the right, or up.

3.9.7.2 Detent controls. Detent controls shall be selected whenever the operational mode requires control operation in discreet steps.

3.9.7.3 Feedback. Feedback of control response adequacy shall be provided as rapidly as possible. Critical control functions shall provide adequate feedback to the operator to ensure that the entry is, in fact, the one that the operator desired to enter.

3.9.7.4 Prevention of accidental actuation. Controls shall be designed and located so that they are not susceptible to being moved accidentally, particularly controls whose inadvertent operation might cause damage to equipment, injury to personnel or degradation of system functions. Cycling through the control's "ON/OFF" position shall be avoided.

3.9.7.5 Positive indication. An indication of control activation shall be provided (e.g., snap feel, audible click, or indicator light illumination).

3.10 Ducts and air openings.

3.10.1 Ducts and duct assemblies. The heater shall be provided with two 15-foot long ducts. The ducts shall conform to the requirements in Appendix A, and to AIA/NAS 1369, type A, form C, except where noted in Appendix A. The duct construction shall conform to AIA/NAS 1370. During abusive handling, twisting, and bending, a minimum amount of damage and crimping of fabric shall occur. The duct fasteners/connectors shall be permanent positive screw clamps or tension latch type and shall prevent separation or leakage from vibration or repositioning of the ducts. The duct color shall be camouflage green. Color standard samples are available (see 4.4.1 and 6.7).

3.10.1.1 Storage devices. When specified (see 6.2), the duct assemblies shall be furnished with a built-in storage device. The built-in storage device shall be internally located in each duct assembly to act as a guide in expanding and retracting the duct assembly and to firmly retain it in the storage position. The storage device shall not damage the fabric during the expansion or retraction of the duct assembly. The carrier portion shall be firmly attached to the male connector.

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3.10.2 Air openings. All external openings shall be protected with a corrosion resistant wire screen and captive covers. All openings except the combustion air intake shall have compatible mating means for connecting to the ducts as specified herein.

3.10.3 Air circulation. The fresh, re-circulation and heated air passageways, including the ducts and connections, shall provide a sealed system. The system (excluding the inlet duct) shall maintain a positive pressure when operated in the air re-circulation mode with the heater against 0.5-inch water gauge pressure.

3.10.4 Fresh makeup air. The heater shall be capable of delivering fresh makeup air up to 5 percent of the total heater airflow rate established by the performance test (see 4.6.3). The fresh makeup air inlet shall be equipped with an adjustable damper. When placed in the fully closed position, the damper shall provide a positive seal against intake of contaminated air. The ventilating air shall be distributed or mixed so that the temperature of the air leaving the heater shall not vary more than 30 °F from the discharge temperature specified in 3.3.1.

3.11 Workmanship.

3.11.1 Common parts. Interchangeable hardware, fastening devices, and a minimum number of types and sizes of bolts, screws, nuts, washers, and similar common parts are desired.

3.11.2 Treatment and painting. Unless otherwise specified (see 6.2), the exterior color of the heater shall be Camouflage Green. Externally exposed surfaces of the heater shall be cleaned, treated and painted unless otherwise specified herein. Unless otherwise specified (see 6.2), the paint shall be a chemical agent resistant coating (CARC). No deterioration, corrosion or damage to paint or treatment of externally exposed surfaces shall be observed. Exterior components shall contribute to field camouflage.

3.11.2.1 CARC paint system. The contractor shall use a CARC paint system from an Army Research Laboratory qualified vendor. A list of qualified vendors can be obtained by contacting the US Army Research Laboratory, Weapons and Materials Directorate, ATTN: AMSRL-WM-M, Aberdeen Proving Ground, MD 21005-5069. Prior to painting any production units, the contractor shall provide documentation showing that the CARC paint has been certified by the Army Research Laboratory.

3.11.2.2 Unpainted surfaces. No deterioration, corrosion or damage to treatment of unpainted surfaces shall be observed. All sheet metal interior to the heater, including those behind insulation materiel, shall be cleaned and treated, but shall not be painted.

3.11.2.3 Camouflage pattern data. When specified (see 6.2), the contractor shall provide camouflage pattern data.

3.11.2.4 Government-furnished-property. When specified (see 6.2), the following property in the quantity indicated will be furnished by the Government (see 6.4):

Description	Quantity
Camouflage paint pattern	As required.

3.11.3 Fastener hardware. Any and all fasteners used in the design and fabrication of this item shall meet the requirements of the applicable ANSI specification. Externally threaded fasteners (e.g., bolts) shall also meet

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the requirements of SAE J429. Bolt holes shall have the burrs removed. All fasteners shall have full thread engagement. The upset rivet heads shall be in accordance with SAE J492. Sheet metal or self-tapping screws shall not be used. Appropriate means shall be used for quick opening access required for operating adjustments and Preventive Maintenance Checks and Services (PMCS).

3.11.4 Gaskets and seals. All gaskets, synthetic rubber seals, etc., shall be suitable for use with fuels specified in 3.6.1, temperatures from -65 to +200 °F, and in all environmental conditions specified herein.

3.11.5 Metal fabrication. Metal used in fabrication shall be free from kinks. Sharp edges; sharp bends and corners shall be square and true and fabricated to minimize surfaces that would trap contaminants or decontaminants. All bends shall be made by controlled means to ensure uniformity of size and shape. All exposed edges and corners shall be rounded to a minimum of 0.03-inch radius. Sharp edges and corners that present a personal safety hazard or potential damage to equipment during usage shall be suitably protected or rounded to a minimum radius of 0.5-inch.

3.11.6 Electrical wiring. The electrical wiring design, protection, methods and materials shall meet the environmental and performance requirements specified herein and shall conform to the National Electric Code 70 (NFPA 70). Cables and wires shall be secured with mechanically mounted cable clamps to ensure correct routing, so that cables or wires do not hinder or obstruct equipment maintenance, and to prevent chaffing from contact with adjacent structures. Fuses shall not be used.

3.11.7 Welding and brazing. All welding shall be in accordance with AWS D1.1, AWS D1.2, or ASME Boiler and Pressure Vessel Code, as applicable. The contractor shall obtain certification that each welder or welding operator is certified in accordance with AWS D1.1, AWS D1.2, AWS B2.1, or ASME Boiler and Pressure Vessel Code, as applicable. Certification shall be made available for review by the contracting officer or designated representative. Brazing of steels, copper alloys and nickel shall conform to AWS B2.2. During the brazing operation, care shall be exercised to protect all components from deformation and damage. Heavy beads and fillets of braze material shall be avoided.

3.11.8 Accessibility. All working parts of the heater sub-assemblies shall be accessible through convenient doors or removable panels.

3.11.9 Bearing lubrication. Bearings in the motor, fuel pump, and fan shall be sealed and permanently lubricated for operation in an ambient temperature range of -65 to +250 °F. If compatible with all other requirements, a fuel pump design utilizing fuel for lubrication is an acceptable alternative for the fuel pump bearing lubrication.

3.11.10 Handbook compartment. A waterproof handbook compartment 9 x 12 x 4 inches shall be provided within the heater enclosure for operating manuals.

3.12 Materials.

3.12.1 Material deterioration prevention and control. The heater shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in applicable storage and operating environments. No deterioration, corrosion or damage to materials or material finishes shall be observed as a result of exposure to test environment conditions or to the fuels specified in 3.6.1. All steel fasteners, screws,

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pins, bolts and common parts shall be treated to provide protection against corrosion.

3.12.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion.

3.12.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for all materials used in the heater design, and shall make this information available upon request to the contracting officer or designated representative.

3.12.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the heater may be newly fabricated from recovered materials to the maximum extent practicable, provided the heater produced meets all other requirements specified herein. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the heater.

3.13 Safety. The heater shall be designed so that under all conditions of normal use (installation, operation, maintenance) and under a likely fault condition (including human error), it protects against the risk of electric shock and other hazards. All hazards shall be eliminated or reduced to the lowest risk level practicable using methods in the following order of precedence: design; incorporation of safety devices; incorporation of warning devices; and procedures/training. Catastrophic or critical hazards shall not rely solely on warnings, cautions, or procedures/training for control of risk. All safety hazards not eliminated through design shall be addressed in the appropriate technical manuals. Information regarding hazard-avoiding procedures and safety warning labels on equipment shall be included in all manuals. Maintenance technical manuals shall address replacement procedures for damaged or missing safety labels. The heater shall comply with SEL Form 1183 (see 6.2), which shall be appended to the Safety Assessment Report.

3.13.1 Electrical safety. The following shall be incorporated into the heater design:

a. Operators shall not have access to components with voltages exceeding 30 volts (V). The operator shall not be exposed to stored energy shock at the disconnecting means.

b. Protection shall be provided to personnel during maintenance and repair to prevent unintentional contact with voltages exceeding 70V. Current sources exceeding 25 amps (A) shall be protected from accidental short circuiting. Capacitors shall be discharged to less than 30V and 20 joules (J) energy prior to maintainer access.

c. The heater shall be protected from damage when connected to incorrect input power/voltage levels or polarity. Equipment shall be designed so that all external parts will be at ground potential.

d. All multi-pin connectors shall be keyed to prevent incorrect mating or insertion. All electrically hot contacts shall be socket contacts. All protective covers shall be captive.

3.13.2 Mechanical safety. The following shall be incorporated into the heater design:

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a. The equipment shall provide maximum access and safety to personnel during installation, operation, and maintenance.
b. The heater shall be designed so that it can be handled and lifted safely.
c. Pressurized components shall be provided with blowouts or relief valves which vent in a safe location and manner.
d. All rotating or reciprocating parts and parts subject to high operational temperatures that are of such nature or so located as to be a hazard to operating personnel shall be guarded or insulated to the extent necessary to eliminate the hazard.
e. Fuel drains shall not allow fuel accumulation, spills, or overflows to run onto hot surfaces or electrical equipment.
f. Exhausts shall be directed so that they do not endanger personnel.
g. The heater design shall not permit any momentary body contact with any surface that has a temperature greater than +140 °F. The design shall not permit any prolonged body contact with any metal surface that has a surface temperature of more than +120 °F, or any plastic surface that has a surface temperature of +140 °F.

3.13.3 Fuel system safety. Components of the fuel system shall be routed, located or guarded so as to be protected from mechanical damage and extreme temperatures. The appropriate fuel types for heater operation shall be clearly labeled at the fuel fill port.

3.13.4 Material safety. Materials used in the heater shall be fire retardant and shall not emit noxious or toxic fumes during operation. Toxic materials contained within the heater shall be controlled so that they do not present a hazard to the user under any condition. Asbestos materials shall not be used.

3.13.5 Safety markings and labels. Safety markings and labels shall be provided identifying any potential hazards to personnel. Safety markings and labels shall comply ANSI Z535.4, UL 1950 or UL 969. Voltages in excess of 70V shall use the signal word "WARNING". Safety labels shall not be removed when a barrier or access door is opened/removed. Colors of safety critical controls and indicators shall be yellow for caution, and red for danger.

3.14 Reliability, endurance and maintainability.

3.14.1 Reliability. The heater shall demonstrate a Mean-Time-Between-Failure (MTBF) of 450 hours with at least 85 percent confidence.

3.14.2 Endurance. The heater shall have a minimum endurance of 2,000 hours before a durability failure (see 6.3.3).

3.14.3 Maintenance ratio. The heater shall demonstrate a minimum maintenance ratio (MR) of 0.042 mmh/operating hour (maintenance man-hour/operating hour). Maintenance man-hours include man-hours expended for scheduled and unscheduled maintenance before, during and after services, including time spent for inspection, diagnosis, and adjustment of heater and repair of failed components and assemblies. Preventive maintenance checks and services performed by the operator under 20 minutes are excluded.

3.15 Maintenance requirements.

3.15.1 Module replacement. The heater shall be repairable by module replacement by operator/crew (unit) level personnel.

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3.15.2 Malfunction identification. Equipment design shall facilitate rapid and positive fault detection and isolation of defective items to permit prompt removal and replacement. Visual warning devices shall be provided to indicate overcurrent, over/under voltage, high pressure, low pressure, and heater overtemperature malfunctions that could cause severe injury/equipment damage. Visual indicators shall be located on the remote control box, readily visible, and labeled to indicate the appropriate fault condition as specified herein.

3.15.3 Preventive maintenance checks and services (PMCS). PMCS shall be 20 minutes per 8-hour shift and a desired PMCS of 20 minutes per day.

3.15.4 Scheduled maintenance. Scheduled maintenance services shall not be required more frequently than each 250 hours of operation. Scheduled maintenance shall not be required at any level higher than operational.

3.15.5 Relative accessibility. Mission critical items which require rapid maintenance shall be most accessible. When relative criticality is not a factor, items requiring most frequent access shall be most accessible. High failure rate items shall be accessible for replacement without moving non-failed items.

3.15.6 Standard tools and equipment. The heater shall require use of only those tools in the general mechanics kit specified in Supply Catalog SC 5180-90-CL-N26. Test, measurement, and diagnostic equipment (TMDE) required for maintenance shall currently exist in the DoD supply system. Built-In-Test-Equipment (BITE) for heater malfunction diagnosis shall be incorporated in the design where practicable.

3.15.7 Design for maintainer. The heater shall be designed for ease of maintenance. Standard parts shall be used whenever practicable.

3.15.7.1 Error proof design. Provisions to preclude improper assembly or installation shall include the following:

a. physical measures to preclude interchange of items of a same or similar form that are not in fact functionally interchangeable.
b. physical measures to preclude improper mounting of units or components
c. measures (e.g., coding) to facilitate identification and interchange of interchangeable items. Measures (e.g., alignment pins) to facilitate proper mounting of items
d. measures to ensure that identification, orientation, and alignment provisions include cables and connectors.

3.15.7.2 Captive fasteners. Captive fasteners shall be used whenever practicable.

3.16 Marking and labeling. No deterioration, corrosion or damage to any marking, identification marking, labeling, instruction plates, schematics, wiring diagrams, or the plates and fasteners utilized for marking shall be observed. Labels shall be mechanically affixed.

3.16.1 General.

a. Marking and labeling shall be as permanent as the normal life expectancy of the marked item, and shall be capable of withstanding the environmental tests and cleaning procedures for the marked item. Labels shall be clear and distinct; all be mounted so as to minimize wear or illegibility due to grease, grime, or dirt; and shall remain legible for the overhaul interval of the equipment on which they are mounted.

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b. All markings shall be capital letters with letter height of at least 0.16-inch, except as specified herein, or shall be easily interpretable icons. Numerals shall be Arabic, except when Roman numerals are used for type designation in specifications or standards. Text type shall be Gothic No. 2 or equivalent.

c. Variable controls, such as temperature controls, shall be labeled with a symbol for a continuously variable control conforming to SAE J1362, symbol 8.7.3.

d. Marking materials creating hazardous conditions shall not be used.

e. Labels and information thereon shall be oriented horizontally so that they may be read quickly and easily from left to right.

f. Labels shall be placed on or very near the items which they identify so as to eliminate confusion with other items and labels. Labels shall be located to avoid obscuring other information needed by the operator. Controls shall not obscure labels.

g. Labels and placards shall be designed to be read easily and accurately at the anticipated operational reading distances, vibration or motion environment, and illumination.

3.16.2 Information plates. The following information plates shall be mechanically affixed to the heater housing in a location where the plates will be both visible and legible: identification plate, instruction plates, electrical schematic, wiring diagram, and fuel system diagram. All marking on these information plates shall be 0.25-inch ± 0.063 high.

3.16.2.1 Identification plate. The contractor shall provide an identification plate for the heater. It shall conform to A-A-50271, composition A, B, or C. The marking color shall be white or silver on a black background. It shall include the following information:

a. serial number
b. manufacturer's name
c. date of manufacture (month/year)
d. contract number
e. National Stock Number
f. item nomenclature (HEATER, ELECTRIC POWERED, MULTIFUEL)
g. model number
h. weight (in pounds)
i. number of this specification

3.16.2.2 Instruction plates. Instruction plates shall be provided which describe clear and concise instructions for set-up and operating instructions for the heater in all modes. Instruction plates shall be readable by the operator at the typical operating position. These plates shall conform to A-A-50271, composition C.

3.16.2.3 Electrical schematic, fuel system diagram, and wiring diagram. An electrical schematic, wiring diagram, and fuel system diagram shall be photoetched on plates conforming to A-A-50271, composition C. Identification markings of control devices and connections shall be shown on all diagrams and shall coincide with markings on all items which are identified in the applicable schematic or wiring diagram.

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3.16.3 Items requiring labeling. At a minimum, the following items in the heater shall be labeled:

3.16.3.1 Electrical components. At a minimum, electrical components shall be labeled with identification marking correlating to the electrical schematic and wiring diagram. They shall also be labeled with the component's vendor information and electrical characteristics to the extent practicable.

3.16.3.2 Fuel system components. At a minimum, fuel system components shall be labeled with identification marking correlating to the fuel system diagram. They shall also be labeled with the component's vendor information and performance characteristics to the extent practicable.

3.16.3.3 Altered items. The contractor shall label the altered items with, at a minimum, a contractor-unique part number for use in obtaining spare and repair parts. The item shall also be labeled with the component's mechanical and electrical characteristics to the extent practical. The item vendor's information shall not be removed.

3.16.3.4 Operator/Maintainer controls. Labels shall be provided to identify operator or maintainer related controls, displays, access covers, and cable connectors. Controls shall be labeled to indicate functional result of control movement (e.g., "ON", "OFF", "INCREASE", etc.).

3.16.3.5 Other parts/components. Other repairable and replaceable components shall be labeled with unique part numbers for use in obtaining spare and repair parts.

3.16.4 Label colors. The label color shall be either white text on a black background, or black text on a green background that is identical to the heater exterior color.

3.17 Human performance and human factors engineering (HFE). The heater shall comply with the following human performance and HFE criteria:

3.17.1 General. The heater shall be operable and maintainable by 5th through 95th percentile soldiers dressed in the Battle Dress Uniform (BDU), cold-wet weather protective clothing, arctic clothing, and the protective ensembles for Mission Oriented Protective Posture (MOPP) Level IV. The heater shall be started and operated by a single operator.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Quality conformance inspection (see 4.3).

4.2 First article inspection.

4.2.1 First article examination. The first article units shall be examined in accordance with table I. Presence of one and more defects shall be cause for rejection of the unit.

4.2.2 First article tests. The contractor shall conduct tests marked "X" in column 1 of table II on the first article units. Failure of any test shall be cause for rejection of the unit.

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4.2.3 Preliminary run-in. Any preliminary run-in of the heater, as may be recommended by the contractor, shall be performed by the contractor prior to submission of the heater for first article tests.

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for tests shall be in accordance with ANSI/ASQC Z1.4.

4.3.2 Examination. Each individual unit shall be examined as specified by an asterisk (*) in table I. Presence of one or more defects shall be cause for rejection.

4.3.3 Tests.

4.3.3.1 Individual. Each unit shall be subjected to tests marked "X" in column 2 of table II. Failure of any test shall be cause for rejection of the unit.

4.3.3.2 Lot. Samples selected in accordance with 4.3.1 shall be subjected to tests marked "X" in column 3 of table II. Failure of any test shall be cause for rejection of the lot.

4.4 Examination schedule.

TABLE I. Examination schedule.

NUMBER	DEFECT	REQUIREMENT
*101.	Dimensions not as specified	3.3.2
*102.	Weight not as specified.	3.3.3
*103.	Transportability provisions not as specified.	3.7 - 3.7.2
104.	Mobility and parking devices not as specified.	3.7.5
*105.	Electric power cable, lead assembly and connectors not as specified.	3.5.2, 3.5.3
106.	Fuel system not as specified.	3.6
107.	Ducts, couplings, connectors, screens and covers not as specified.	3.10 - 3.10.4
108.	Combustion system not as specified.	3.8 - 3.8.8
109.	Controls, switches, indicators not as specified.	3.9 - 3.9.7.5
*110.	Workmanship and manufacturing requirements not as specified.	3.11
111.	Treatment and painting not as specified.	3.11.2 - 3.11.2.4
112.	Topcoat color not as specified.	3.11.2
113.	Camouflage pattern data not provided as specified.	3.11.2.3
*114.	Handbook compartment not provided.	3.11.10
*115.	Materials not as specified.	3.12
116.	Materials not resistant to corrosion and deterioration or treated to be made resistant to corrosion and deterioration of the applicable storage and operating environments.	3.12.1
117.	Instruction and information plates not provided as specified.	3.16.2.2
118.	Contractor does not have documentation available for identification of materials, material finishes, or treatment.	3.12.3

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NUMBER	DEFECT	REQUIREMENT
119.	Used, rebuilt or remanufactured components, pieces, or parts incorporated in the heater.	3.12.4
120.	Material safety not as specified.	3.13.4
121.	Safety design not as specified.	3.13 - 3.13.5
122.	Marking and labeling not as specified.	3.16
123.	Items not labeled as specified.	3.16.3
124.	Human factors engineering design not as specified.	3.17

* Each individual unit shall be examined.

4.4.1 Verification using standard samples. Use standard samples to verify colors with visual inspections. Compare with the naked eye in simulated daylight.

4.5 Test schedule and conditions. Unless otherwise specified herein, the following conditions shall apply:

a. Tests shall be conducted using 120 V, single phase, 50/60 Hz input electrical supply.
b. Tests specified in table II shall be conducted at standard ambient temperature conditions between -40 and +65 °F with all components and appropriate accessories in place.
c. Tests shall be conducted using heaters with the exact same configuration as of the production units.
d. Tests shall be conducted in the sequence shown in table II.
e. All electrical cables with connectors and instrumentation with related materials (i.e., thermocouple wire, test leads, etc.), shall be provided for each test specified herein.
f. No braces, insulating or sealing materials, tiedown devices, or other items that are not inherently a part of the heater shall be employed.
g. Components subject to atmospheric corrosion shall not be treated with any treatment such as grease or oil to meet test requirements.
h. Unless otherwise specified, test cycles are 24-hour periods.
i. Tests shall be conducted using JP-8 fuel.

TABLE II. Test schedule.

First Article Test Unit No.						Quality Conformance		Test	Test Paragraph	Requirement Paragraph
1						2	3	4	5	6
1	2	3	4	5	6	IND	LOT			
X	X	X	X	X	X			Examination	4.2.1	4.4
X	X	X	X	X	X	X		Operational Test	4.6.2	3.9.2
							X	Burn-in	4.6.2.1	3.9.2
X	X							Performance	4.6.3	3.3.1
X								High temperature	4.6.4.1	3.4.1
X	X							Low temperature	4.6.4.2	3.4.1, 3.4.2, 3.5, and 3.17.1
	X							Humidity	4.6.4.3	3.4.3, 3.12.1, and 3.4.5
	X							Salt fog	4.6.4.4	3.4.3, 3.12.1
	X							Rain test	4.6.4.5	3.4.4, 3.12.1
	X							Fungus	4.6.4.6	3.4.5
X								Altitude	4.6.4.7	3.3..1, 3.4.1,

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First Article Test Unit No.						Quality Conformance		Test	Test Paragraph	Requirement Paragraph
1						2	3	4	5	6
1	2	3	4	5	6	IND	LOT			
										3.8.2
X								Slinging and Tiedown	4.6.5	3.7, 3.7.1, and 3.7.2
X	X	•			•		X	Noise limits	4.6.6	3.4.6
X	X							Multifuel capability	4.6.7	3.6.1
X								EMI	4.6.8	3.5.1
X								Inclined operation	4.6.9	3.7.6
X								Emissions	4.6.4.7	3.8.2
X	X					X		External power	4.6.10	3.4.2, 3.5
X	X		Δ	Δ			X	Heated air quality	4.6.11	3.8.7
X	X						X	Controls	4.6.12	3.9
X	X						X	Thermostat	4.6.12.1	3.9.5
X	X						X	Flame failure safety	4.6.12.2	3.8.4, 3.9.3
X	X						X	High temp safety	4.6.12.3	3.8.5, 3.9.4
X								Rail impact	4.6.13	3.7.4
X								Vibration	4.6.14	3.7.3
		X	X	X	X			REM	4.6.15	3.14

Δ See 4.6.15.4

• See 4.6.15.5

4.6 Test procedures.

4.6.1 Functional test. The heater shall be operated in each of its operational modes with the temperature selection control set to the warmest and then to the coolest setting for 10 minutes per temperature control setting, where applicable. For example: Run #1 is a 10-minute operation at "VENT"; Run #2 is a 10-minute operation at "AUTO Heat"-coolest; Run #3 is a 10-minute operation at "AUTO heat"-warmest; etc. Record all of the data for each test run.

4.6.2 Operational test. The heaters shall be started in less than one minute at a temperature between -40 and +65 °F. Operate the heater for one hour in each of the three modes specified in 3.9.2. The heater shall be started and operated in such a manner as to demonstrate satisfactory operation of all components, and all controls used in the entire heater assembly. The failure of any component, control, or accessory to function properly shall constitute failure of this test.

4.6.2.1 Burn-in. One unit of the sample, selected in accordance with 4.3.1, shall be operated for 45 hours. Operate the heater for one hour in each mode specified in 3.9.2 and repeat the operation 15 times. The failure of any component, control, or accessory to function properly shall be cause for rejection of the lot.

4.6.3 Performance test. The test shall be conducted in accordance with ASHRAE 45 air throughput method with an air inlet temperature of 0 ±2.5 °F at sea level atmospheric pressure. Data shall be collected to determine the temperature at the output duct, airflow, heat output, efficiency, and heat

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content of the fuel as specified in A-A-52557. The test shall be conducted with the heaters operating on grade Low Sulfur No. 1-D diesel fuel. Failure to meet the requirements specified in 3.3.1 shall constitute failure of this test.

4.6.4 Environmental testing. Each environmental subtest specified herein shall be followed by an operational test as specified in 4.6.2 to determine that the functional ability of the heater and controls has not been impaired by the environmental exposure. A failure of any operational test shall constitute failure of the preceding environmental subtests.

4.6.4.1 High temperature storage. The heater shall be subjected to a high temperature storage test as specified herein, for three 24-hour cycles at the environmental conditions described in table III.

TABLE III. High temperature cycles.

Time	Temperature °F	RH* %	Time	Temperature °F	RH %
0100	91	36	1300	142	6
0200	90	38	1400	114	6
0300	90	41	1500	145	5
0400	88	44	1600	144	6
0500	86	44	1700	140	6
0600	88	43	1800	134	6
0700	93	32	1900	122	10
0800	101	30	2000	111	14
0900	107	23	2100	101	19
1000	113	17	2200	95	25
1100	124	14	2300	93	28
1200	134	8	2400	91	33

RH - relative humidity

4.6.4.1.1 Test item installation. The heater shall be placed in the test facility with a minimum of 6 inches of clearance between the heater, all ducts installed on the heater, and the walls and ceiling, to allow proper air circulation.

4.6.4.1.2 Test item configuration. All access panels shall be in place. The appropriate cover/cap shall be in place on each unused electrical connector.

4.6.4.1.3 Instrumentation. All test instrumentation shall meet the measuring tolerances specified in ASHRAE Standard 45. The heater shall contain instrumentation to accurately monitor and record all of the data specified in 4.6.4.1.5. Thermocouple grids which yield an average temperature reading over the grid area shall be mounted on each supply and return air opening. The test facility shall have instrumentation capable of maintaining and continuously monitoring the required conditions of temperature and humidity.

4.6.4.1.4 Test condition tolerances. Tolerances for test conditions shall be as follows:

a. Input power frequency: ±1 percent.
b. Input power voltage: ±1 percent.

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c. Temperature: ±3 °F.
d. Elapsed time shall be measured with an accuracy of ±1 percent.
e. Relative humidity at the chamber control sensor shall be ±5 percent of the measured value.

4.6.4.1.5 Data. The test unit nomenclature and serial number, the test being performed, and the test run number shall be recorded on each data sheet. In addition, the following shall be monitored and recorded at 10-minute intervals throughout the test and stabilization periods to determine that normal heater operation is being maintained and to identify problems in event of failure or malfunction:

a. Voltage (Volts).
b. Current (Amps).
c. Frequency (Hertz).
d. Test chamber ambient air temperature (°F)
e. Test chamber relative humidity (percent RH)
f. Air temperatures (°F).
(1) Return air duct (dry bulb).
(2) Supply air duct (dry bulb).

In addition the following data shall be recorded as required during each test:
a. Any failures.
b. Elapsed time prior to failure.
c. Corrective action taken to correct failure.

4.6.4.1.6 Procedure. The test procedure is as follows:

Step 1. Install the required temperature sensors on, in, and around the heater and test facility.

Step 2. Place the heater and all the associated equipment, including the duct assembly, in the environmental chamber in an unprotected configuration (no shipping/storage container).

Step 3. Set and stabilize the chamber environment to the initial test conditions specified in table III for time 0000 of the specified cycle. Stabilization shall be considered to be reached after three consecutive test chamber temperature readings are equal within 3 °F.

Step 4. After stabilization has occurred, the heater shall be subjected to the temperature and humidity conditions of the storage cycle for three 24-hour cycles.

Step 5. At the completion of the last cycle, adjust the chamber air temperature to controlled ambient conditions and maintain until temperature stabilization of the heater has been achieved.

Step 6. Conduct a complete visual inspection of the heater and a operational test (see 4.6.2) and record the results.

4.6.4.1.7 Test interruption. In the event of test interruption, the contractor shall note the cause of the interruption and the time into test of the interruption. If the interruption was caused by a heater failure, the test shall be considered failed. If the interruption is caused by some other reason that causes the test conditions to fall out of allowable tolerances, the test must be restarted at the end of the last successfully completed 24-hour cycle.

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4.6.4.1.8 Failure criteria. The following shall constitute failure of this test:

a. Nonconformance to 3.4.1.
b. Any evidence of damage, corrosion or deterioration.
c. Failure of the heater to start and operate.

4.6.4.2 Low temperature storage. The heater shall be subjected to low temperature storage test at −60 °F ambient temperature and manipulation and operation tests at −40 °F.

4.6.4.2.1 Test item installation. The heater shall be placed in the test facility with a minimum of 6 inches of clearance between the heater, and all ducts installed on the heater, and the walls and ceiling to allow proper air circulation.

4.6.4.2.2 Instrumentation. All test instrumentation shall meet the measuring tolerances specified in ASHRAE Standard 45. The heater shall be instrumented to accurately monitor and record all of the data specified in 4.6.4.2.4. Thermocouple grids which yield an average temperature reading over the grid area shall be mounted on each supply and return air opening. The test facility shall have instrumentation capable of maintaining and continuously monitoring the required conditions of low temperature.

4.6.4.2.3 Test condition tolerances. Tolerances for test conditions shall be as follows:

a. Input power frequency: ±1 percent
b. Input power voltage: ±1 percent
c. Temperature: + 0, −10 °F
d. Pressure (including air, barometric): ±5 percent
e. Elapsed time shall be measured with an accuracy of ±1 percent.

4.6.4.2.4 Data. Collect data as specified in 4.6.4.1.5.

4.6.4.2.5 Test procedure.

Step 1. Install the required temperature sensors on, in, and around the heater and test facility.

Step 2. Place the heater and all the associated equipment, including the duct assembly, in the environmental chamber in an unprotected configuration (no shipping/storage container) and stabilize at ambient conditions (see 4.6).

Step 3. Conduct a complete visual examination on the heater with special attention to stress areas such as corners, wires, insulating material. Document the results.

Step 4. Conduct an operational test (see 4.6.2) on the heater at ambient temperature conditions. Document the results.

Step 5. Adjust and stabilize the test chamber to the storage temperature of −60 °F.

Step 6. Soak the heater (packed with accessory equipment, doors secured and uncrated) for a minimum of 24 hours at a temperature of −60 °F.

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Step 7. Conduct a visual examination of the heater and compare the results with the pretest data. Record physical changes, if any.

Step 8. Adjust and stabilize the chamber temperature to -40 °F (maximum cool-down rate shall not exceed 6 °F per minute).

Step 9. With personnel clothed and equipped as they would be in a low-temperature tactical situation, unpack, set up for operation and repack the heater and its accessories at a temperature of -40 °F.

Step 10. Two heaters shall be cold soaked at an ambient temperature of -40 °F for a period of 12 hours.

Step 11. Following the cold soak, with the temperature maintained at -40 °F, operate one heater in "Manual" heat and the other heater in "Auto" heat mode for 1 hour. The test shall be conducted with both heaters operating on JP-8.

Step 12. Conduct the external power test (as specified in 4.6.10) at -40 °F on one of the heaters in all three modes ("VENT", "MANUAL", "AUTO").

4.6.4.2.6 Test interruption. In the event of test interruption, the contractor shall note the cause of the interruption and the time into test of the interruption. If the interruption was caused by an heater failure, the test shall be considered failed. If the interruption is caused by some other reason, the entire test procedure shall be repeated.

4.6.4.2.7 Failure criteria. The following shall constitute failure of this test:

a. Nonconformance to 3.4.1, 3.4.2, 3.5, and 3.17.1.
b. Any evidence of damage, corrosion or deterioration.
c. Failure of the heater to start and operate.

4.6.4.3 Humidity. The heater shall be placed in the test facility with a minimum of 6 inches of clearance between the heater, and all ducts installed on the heater, and the walls and ceiling to allow proper air circulation.

4.6.4.3.1 Test conditions. The heater shall be placed in the test facility with a minimum of 6 inches of clearance between the heater, and all ducts installed on the heater, and the walls and ceiling to allow proper air circulation. The heater shall be subjected to the hot-humid cycle in table IV for 20 cycles. Operational tests (see 4.6.2) shall be performed at least once every 5 days at the existing temperature-humidity conditions. A visual inspection and functional test shall be conducted after 7 days exposure to hot humid cycle conditions.

4.6.4.3.2 Test facility. The test facility shall be as follows:

a. The required apparatus shall consist of a chamber or cabinet, and auxiliary instrumentation capable of maintaining and continuously monitoring the required conditions of temperature and relative humidity throughout an envelope of air surrounding the heater.

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b. Unless otherwise specified, the test volume of the chamber and the accessories contained therein shall be constructed and arranged in such a manner as to prevent condensate from dripping on the heater. The test volume shall be vented to the atmosphere to prevent buildup of total pressure and prevent contamination from entering. Relative humidity shall be determined by employing either solid-state sensors whose calibration is not affected by water condensation or by an equivalent method, such as fast-reacting wet-bulb/dry-bulb sensors or dewpoint indicators. Sensors that are sensitive to condensation, such as the lithium-chloride type are not recommended for tests with high relative humidity levels. A data collection system separate from the chamber controllers shall be employed to measure test volume conditions. A recording device shall be mandatory for the data collection system. If charts are used, the charts shall be readable within ±3 °F. If the wet-wick control method is approved for use, the wet bulb and tank shall be cleaned and a new wick installed before each test and at least every 30 days. Water used in wet-wick systems shall be of the same quality as that used to produce the humidity. Water bottle, wick, sensor and other components making up relative humidity measuring systems shall, when physically possible, be visually examined at least once every 24 hours during the test. The velocity of air flowing across the wet-bulb sensor shall be not less than 900 ft/min (4.5 m/s), and the wet wick shall be located on the suction side of the fan to eliminate the effect of fan heat. The flow of air anywhere within the envelope of air surrounding the heater shall be maintained between 98 and 335 ft/min (0.5 to 1.7 m/s).

c. Relative humidity within the envelope of air surrounding the heater shall be created by steam or water injection. Water used in either method shall be distilled, demineralized, or de-ionized and have a resistivity of not less than 250,000 ohm-centimeters at 77 °F (25 °C). Its quality shall be determined at periodic intervals (not to exceed 15 days) to ensure its acceptance. If water injection is used to humidify the envelope of air, the water shall be temperature conditioned before its injection to prevent upset of the test conditions and shall not be injected directly into the test section. Condensation developed within the chamber test volume during the test shall be drained from the test volume and discarded.

d. No material that will cause the heater to deteriorate or that will affect the test results other than water shall be brought into physical contact with the heater. No rust or corrosive contaminants or any material other than water shall be introduced into the chamber test volume.

e. Dehumidification, humidification, heating, and cooling of the air envelope surrounding the heater shall be achieved by methods that do not change the chemical composition of the air, water, or water vapor within that volume of air.

4.6.4.3.3 Controls. Temperature and relative humidity measurements made during the test shall be continuous if in analog form, or at intervals of 15 minutes or less if in digital form. Tolerances for test conditions shall be as follows:

a. Input power frequency: ±1 percent
b. Input power voltage: ±1 percent
c. Temperature: ±3 °F
d. Elapsed time shall be measured with an accuracy of ±1 percent
e. Relative humidity at the chamber control sensor shall be ±5 percent of the measured value.

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4.6.4.3.4 Test interruption.

a. An undertest interruption shall be best handled by keeping the chamber closed in an effort to maintain tolerances. As long as the tolerances are maintained, re-testing shall be resumed by re-establishing the prescribed conditions and continuing from the point of the interruption. If an unscheduled interruption occurs that causes the test conditions to exceed the allowable tolerances toward standard ambient temperatures, the test shall be reinitiated at the end of the last successfully completed cycle. Any test item failure that occurs shall be treated as a failure.

b. An interruption that results in exposure of the heater to conditions more extreme than specified herein shall be followed by a complete physical examination and performance of a functional test (see 4.6.1) of the heater before continuation of testing. When a problem is discovered, the preferable course of action shall be to terminate the test and reinitiate testing with a new heater. If no problem has been encountered, pre-interruption conditions shall be re-established and continue from the point where the test tolerances were exceeded.

4.6.4.3.5 Pretest standard ambient checkout. The following checkout shall be performed:

a. Proper operation of the test chamber shall be verified prior to testing.

b. The heater shall be inserted into the chamber.

c. Plugs, covers, and access plates not used in operation, but used in servicing shall remain in place.

d. The chamber shall be adjusted to the following controlled ambient conditions and maintained for 24 hours:

(1) Temperature 73 ±3.6 °F
(2) Relative humidity 50 ±5 percent.
(3) Atmospheric pressure 28.5 +2.0, - 3.0 inches Hg.

e. A complete visual examination of the heater shall be conducted and the results recorded.

f. A functional test as specified in 4.6.1 shall be performed and the results recorded.

4.6.4.3.6 Procedure. The following procedure shall be used for the humidity test:

a. With the heater in the chamber, the chamber conditions shall be adjusted to those specified in table IV for time 0000 of the specified cycle.
b. A 24-hour cycle shall be performed with the time temperature humidity values specified in the hot humid cycle in table IV.
c. Functional and operational tests shall be performed as specified in 4.6.4.3.1 at the existing temperature- humidity conditions and the results recorded.
d. Repeat steps b. and c. for the number of cycles indicated in 4.6.4.3.1.
e. At the end of the required number of cycles, the temperature and humidity conditions shall be adjusted to ambient conditions (see 4.6) and maintained for at least 24 hours.

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f. A complete visual examination of the heater shall be conducted, and the results documented.
g. A functional test as specified in 4.6.1 shall be performed and the results recorded.
h. These data shall be compared with the pretest data.

TABLE IV. Humidity test conditions.

Time	Hot Humid Cycle (°F)	Relative Humidity percent	Time	Hot Humid Cycle (°F)	Relative Humidity percent
0000	88	88	1200	104	62
0100	88	88	1300	105	59
0200	88	88	1400	105	59
0300	88	88	1500	105	59
0400	88	88	1600	105	59
0500	88	88	1700	102	65
0600	90	85	1800	99	69
0700	93	80	1900	97	73
0800	96	76	2000	94	79
0900	98	73	2100	91	85
1000	100	69	2200	90	85
1100	102	65	2300	89	88

4.6.4.3.7 Information to be recorded. The following information shall be recorded:

a. Test item identification (manufacturer, serial number, etc.).
b. Previous test methods to which the heater has been subjected.

c. Results of each performance check and visual examination (and photographs if applicable).
(1) Pretest
(2) During test
(3) Post test

d. Length of time required for each performance check.
e. Status of heater for each visual examination.
f. Exposure durations.
g. Procedure and test levels used.
h. Time versus temperature and humidity.

4.6.4.3.8 Failure criteria. Nonconformance to 3.4.3, 3.4.5, or 3.12.1 shall constitute failure of this test.

4.6.4.4 Salt fog test. The heater shall be placed in the test facility with a minimum of 6 inches of clearance between the heater, and all ducts installed on the heater, and the walls and ceiling to allow proper air circulation.

4.6.4.4.1 Test facility. The facility used in performing this test shall include:

a. A test chamber with:

(1) Supporting racks designed and constructed so that they shall not affect the characteristics of the salt fog mist. All parts of the test chamber and the supporting racks that come into contact with the heater shall be constructed of or be buffered with material that will not cause

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electrolytic corrosion. Condensation shall not be permitted to drip onto the heater. No liquid that comes in contact with either the exposure chamber or the heater shall return to the salt solution reservoir. The exposure chamber shall be properly vented to prevent pressure build-up.

(2) The capability to maintain temperatures in the exposure zone at 95 °F (35 °C). The use of immersion heaters within the chamber exposure area for the purpose of maintaining the temperature within the exposure zone shall be prohibited.

b. A salt solution reservoir shall be fabricated from a material that is non-reactive with the salt solution (e.g., glass, hard rubber or plastic).

c. A means of injecting the salt solution into the test chamber shall be provided. Caution shall be exercised to prevent clogging of the nozzles by salt build-up. Atomizers shall be of such design and construction as to produce a finely divided, wet, dense fog. Atomizing nozzles and the piping system shall be fabricated of material that is non-reactive to the salt solution. A filter shall be fabricated of noncorrosive materials similar to that shown in figure 3, and it shall be provided in the supply line as shown in figure 4.

d. A minimum of two salt fog collection receptacles shall be provided. One shall be at the perimeter of the heater nearest to the nozzle, and the other shall be at the same perimeter farthest from the nozzle. Receptacles shall be placed so that they are not shielded by the heater such that they can not collect drops of solution from the heater or other sources.

4.6.4.4.2 Controls.

a. Before injection into the test section, the salt solution shall be heated to within ±10 °F (±6 °C) of the test section temperature at the time of injection.

b. All water used during the test shall be from steam or distilled, de-mineralized, or de-ionized water, and have a pH between 6.5 and 7.2 at 77 °F (25 °C), or have a resistivity of not less than 250,000 ohm centimeters at 77 °F (25 °C).

c. Air velocity in the test chambers shall be essentially zero.

d. The oil and dirt-free compressed air used to produce the atomized solution shall be preheated and pre-humidified such that the temperature is 95 ±3 °F (35 ±1 °C) and the relative humidity is in excess of 85 percent at the nozzle.

4.6.4.4.3 Test interruption.

a. When an unscheduled undertest interruption occurs that causes the test conditions to exceed allowable tolerances toward standard ambient conditions, the heater shall undergo a complete visual inspection, and a technical evaluation shall be made of the impact of the interruption on the test results. The test shall be restarted at the point of interruption, and the heater shall be re-stabilized at the test conditions.

b. When an unscheduled overtest interruption occurs that causes the test conditions to exceed allowable tolerances away from standard ambient conditions, the test conditions shall be stabilized to within tolerances and shall be held at that level until a complete visual inspection and technical

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evaluation can be performed to determine the impact of the interruption on the test results. Pre-interruption conditions shall be reestablished and the test shall be continued from the point of interruption.

4.6.4.4.4 Preparation of salt solution. The salt shall be sodium chloride containing (on a dry basis) not more than 0.1 percent sodium iodide and not more than 0.5 percent total impurities. A 5 ±1 percent solution shall be prepared by dissolving 5 parts by weight of salt in 95 parts by weight of water. The solution shall be adjusted to, and maintained at, a specific gravity (see figure 5) by using the measured temperature and density of the salt solution. Sodium tetraborate (borax) may be added to the salt solution to as a pH stabilization agent in a ratio not to exceed 0.7g sodium tetraborate to 75 liters salt solution. The pH of the salt solution, as collected as fallout in the exposure chamber, shall be maintained between 6.5 and 7.2 with the solution temperature at +95 °F (+35 °C). Only diluted chemically pure hydrochloric acid or chemically pure sodium hydroxide shall be used to adjust the pH. The pH measurement shall be made electrometrically or colorimetrically.

4.6.4.4.5 Chamber operation verification. Immediately before the test, and with the exposure chamber empty, adjust all test parameters to those required for testing. Maintain these conditions for at least one 24-hour period or until proper operation and salt fog collection can be verified. All test parameters shall be continuously monitored to verify proper chamber operation.

4.6.4.4.6 Pretest standard ambient checkout. The checkout shall be conducted at standard ambient conditions (see 4.6) as follows:

a. Place the heater and all the associated equipment, including the duct assembly, in the environmental chamber in an unprotected configuration (no shipping/storage container). Plugs, covers, and access plates not used in operation, but used in servicing shall remain in place.

b. The room ambient conditions shall be recorded.

c. A complete visual examination of the heater shall be conducted with attention to:
   (1) High stress areas.
   (2) Areas where dissimilar metals are in contact.
   (3) Electrical and electronic components.
   (4) Metallic surfaces.
   (5) Enclosed volumes where condensation is likely to occur.
   (6) Components or surfaces provided with corrosion resistant coatings.
   (7) Mechanical components.
   (8) Electrical and thermal insulators.

d. The results of the visual examination shall be documented.

e. A functional test shall be performed as specified in 4.6.1. The results shall be recorded.

f. Any problems shall be recorded and resolved prior to test initiation.

4.6.4.4.7 Preparation of heater.

a. The heater shall be given a minimum of handling, and shall be prepared for test immediately before exposure. The heater shall be free of surface

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contamination i.e., oil, grease, or dirt which could cause a water break. The cleaning methods shall not include the use of corrosive solvents, solvents which deposit either corrosive or protective films, or abrasives other than a paste of pure magnesium oxide.

b. The heater shall be installed when the test chamber is at standard ambient conditions. The heater shall be positioned at least 6 inches from the walls, floor, ceiling, etc., to allow for adequate circulation.

4.6.4.4.8 Aggravated screening procedure.

a. The chamber temperature shall be adjusted to 95 °F (35 °C), and the heater shall be soaked at this temperature for at least two hours prior to introducing the salt fog.

b. Submit the heater to alternating 24-hour periods of salt fog exposure and standard ambient (drying) conditions (see 4.6) for four 24-hour periods (2 wet and 2 dry). During the salt fog exposure periods, the salt solution shall be continuously atomized into the chamber for a period of 24 hours. After each 24-hour salt fog exposure period, the salt fog fallout rate and the pH of the fallout solution shall be measured. Fallout rate shall be between 0.5 and 3.0 ml/80 square cm/hour.

c. After completion of the fourth 24-hour cycle, a functional test shall be performed as specified in 4.6.1. The heater shall be visually inspected, and all results of the inspection and functional test shall be recorded for comparison to the pretest results.

4.6.4.4.9 Information to be recorded. The following information shall be recorded:

a. Heater identification (manufacturer, serial number, etc.).
b. Previous test methods to which the heater was subjected.
c. Results of each visual examination and functional test.
d. Areas of the heater which were visually and functionally examined and an explanation of their inclusion.
e. Areas of the heater not visually or functionally examined and an explanation for their exclusion.
f. Test chamber operational information (interruptions, time schedule, etc.).

g. Test variables
   (1) Salt solution
   (2) Salt solution fallout rate
   (3) Resistivity of initial water and type of water

h. Preliminary failure analysis.

4.6.4.4.10 Failure criteria. The following shall constitute failure of this test:

a. Evidence of serious corrosion of metallic parts or other damage that might affect subsequent operation of the heater or the duct assembly, shall constitute failure of this test.

b. Nonconformance to 3.4.3 or 3.12.1 shall constitute failure of this test.

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4.6.4.5 Rain test. The heater shall be installed in the test facility with a minimum of 6 inches of clearance between the heater, and all ducts installed on the heater, and the walls and ceiling to allow proper air circulation. Rainfall shall be produced by means of water spray and shall be measured with US Weather Bureau type gauges. Water supply temperature shall be between 50 and 70 °F and shall contain a water soluble dye such as fluorescein for ease in determining the locations and extent of leakage. Water from the local water supply shall be used.

4.6.4.5.1 Test facility. The rain facility shall have the capability of producing falling rain accompanied by blowing wind as specified herein. The facility temperature shall be uncontrolled except as regulated by the water introduced as rain. The rain shall be produced by a water distribution system of such design that the water is emitted in the form of droplets having a diameter between 0.5 and 4.5 millimeters. The rain shall be dispersed completely over the heater. The wind source shall be positioned with respect to the heater so that it will cause the rain to beat directly, with variations up to 45 degrees from horizontal, and uniformly against one side of the heater. The wind source shall be capable of producing horizontal wind velocities of no less than 40 miles per hour (mph) and no greater than 45 miles per hour. The wind velocity shall be measured at the position of the heater prior to placement of the heater in the facility. No rust or corrosive contaminants shall be imposed on the heater by the test facility.

4.6.4.5.2 Controls.

a. The rainfall rate shall be verified immediately before each test.
b. The air velocity shall be verified immediately before each test.

4.6.4.5.3 Test interruption. The test shall be normally continued from the point of interruption.

4.6.4.5.4 Pretest standard ambient checkout. A pretest standard ambient checkout shall be performed as follows:

a. The heater shall be stabilized at standard ambient conditions (see 4.6) in the test chamber.
b. A complete visual examination of the heater shall be conducted and the results documented. No sealing materials shall be used. Remove all surface contaminants such as oil, grease or dirt which could cause a water break.
c. Plugs, covers, and access plates not used in operation, but used in servicing shall remain in place.
d. A functional test as specified in 4.6.1 shall be performed and the results recorded.
e. If the heater does not operate properly, the problem(s) shall be resolved and this checkout shall be repeated.

4.6.4.5.5 Procedure. The rain test shall be performed in accordance with the following test procedure:

a. With the heater in the facility and in its normal operating position, adjust the rainfall rate at a minimum of 4 inches per hour.
b. Initiate the wind at a velocity of no less than 40 mph and no greater than 45 mph and maintain for at least 30 minutes.
c. Conduct an operational test as specified in 4.6.2 during the last ten minutes of exposure.

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d. After step c. has been completed, inspect the heater for water intrusion. Any free water found inside the protected areas of the test item shall be measured, documented and then emptied.
e. Rotate the heater to expose to the rain source any other side of the heater not yet exposed to the rain source (top and four sides of the heater).
f. Repeat steps a. through d. until all possible variations have been accomplished.

4.6.4.5.6 Information to be recorded. The following information shall be recorded:

a. Test item identification (manufacturer, serial number, etc.).
b. Previous test methods to which the heater has been subjected.

c. Results of each performance check and visual examination (and photographs if applicable).
(1) Pretest
(2) During test
(3) Post test

d. Length of time required for each performance check.
e. Status of heater for each visual examination.
f. Exposure duration.
g. Rainfall rate.
h. Wind velocity.
i. Water and test item temperatures.
j. Surfaces of the test item subjected to rainfall.

4.6.4.5.7 Failure criteria. One or more of the following shall constitute failure of the rain test:

a. Failure of the heater to start or operate during the last ten minutes of the rain.
b. Malfunctioning of components or an electrical short during operation.
c. Nonconformance to 3.4.4 or 3.12.1.

4.6.4.6 Fungus test. The heater shall be tested in accordance with Appendix B. Test duration shall be 28 days.

4.6.4.6.1 Failure criteria. One or more of the following shall constitute failure of the fungus test:

a. Failure of the heater to start or operate after the fungus test exposure period.
b. Malfunctioning of components or an electrical short during operation.
c. Excessive fungal growth or deterioration of materials or components.
d. Nonconformance to 3.4.5.

4.6.4.7 Altitude. The heater shall be operated at an altitude, either natural or water gauge of 4,000 feet with an ambient temperature between −40 and +65 °F, against a 0.5-inch back pressure. After heater operation has stabilized, operate in the heat mode continuously for a period of 1 hour and record data required to determine the indicated output, indicated efficiency, and flue gas emissions content. The indicated efficiency shall be determined using the method specified in ASHRAE 45, heat balance method, heat balance efficiency. Nonconformance to 3.3.1, 3.4.1, and 3.8.2 shall constitute failure of this test.

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4.6.5 Transportability test.

4.6.5.1 Tiedown provisions test. All tiedown provisions shall be tested attached to the heater. The tiedown provisions shall withstand its proportionate share of the loading as specified in 3.7.2. Loads applied in the longitudinal, vertical, and lateral directions shall be applied statically and independently for not less than 6.0 seconds an shall be not less than the required load in each direction. Loads applied to the provisions shall be measured with an appropriate measuring device, such as a load cell or dynamometer. The following shall constitute failure of this test:

a. Nonconformance to 3.7, 3.7.1 or 3.7.2.
b. Any structural damage to the heater.
c. Weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components.

4.6.5.2 Slinging provisions test. All slinging provisions shall be tested attached to the heater. The heater shall be secured to a fixture which can not be lifted when the load specified herein is applied to it. An overhead crane or winch and a sling with hooks on its free legs both of which are rated to handle the applied load specified herein shall be utilized. A calibrated load cell capable of accurately measuring the load specified herein and of being attached in line between the overhead crane/winch and the sling shall be used. The load cell shall be hung from the overhead crane/winch hook, and the sling shall be hung from the load cell with its free legs dangling downward. The load cell shall be zeroed. Connect the sling hooks to the heater lift provisions. The overhead crane shall be positioned over the center of the heater. Then it shall be raised until a load equal to at least 2.3 times the weight of the heater (as specified in 3.3.3) is applied and maintained for 20 seconds. The heater and lift provisions shall be inspected for structural abnormalities and damage. The following shall constitute failure of this test:

a. Nonconformance to 3.7, 3.7.1 or 3.7.2.
b. Any structural damage to the heater.
c. Failure, deformation or damage to any one of the lift provisions.

4.6.6 Noise.

4.6.6.1 Test site. The test site shall be an either anechoic chamber or an uniform flat grass surface, free of snow, ice or vegetation over 0.15 meters tall; it shall be free of reflecting surfaces such as buildings, trees, or hillsides within 30 meters.

4.6.6.2 Configuration. The heater shall be configured as specified in 4.6 with the ducts installed.

4.6.6.3 Instrumentation.

4.6.6.3.1 Instrument specifications. Sound level meters shall conform to ANSI S1.4, type 1. Band filter sets shall conform to ANSI S1.11, type E, order II. Other noise recording instrumentation or combinations of instrumentation shall conform to SAE J184 and applicable provisions of ANSI S1.4.

4.6.6.3.2 Calibration. Acoustical calibration procedures shall be accomplished in accordance with the sound level meter manufacturer's instructions. Calibrators shall conform to ANSI S1.40. Calibration shall address the influence of microphones, cables, amplifiers and recorders. Calibration shall be performed any tine the instrumentation is changed, before

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and after each test sequence, but not less than at the start and end of each day.

4.6.6.3.3 Microphones. Either pressure microphones having an essentially flat response at grazing incidence (90 degrees) or free field microphones having an essentially flat response at normal incidence (0 degrees) shall be used with the addition of a random incidence corrector. The microphone shall be placed at the measurement location with the sensitive element up.

4.6.6.4 Data. Noise level measurements shall be recorded in units of dB(A). Slow meter damping shall be used. Where the steady-state noise level is 85 dB(A) or greater, the distances and directions from the noise source at which the sound level is equal to 85 dB(A) shall be determined. This 85 dB(A) contour shall be determined from measurements made at positions around the heater at (horizontal) angular increments not greater than 30 degrees, and at the noisiest angle.

4.6.6.5 Background noise. Background noise shall be 10 dB below the test criteria specified in 3.4.6. A windscreen shall be used at wind velocities between 10 and 20 kilometers per hour. Measurements shall not be valid at wind velocities in excess of 20 kilometers per hour.

4.6.6.6 Procedure. Only personnel and equipment essential to performing the test shall be in the test area while measurements are being taken. Noise levels shall be measured with the burner and air mover of the heater operating. The background noise and wind velocity shall be measured and checked for conformance to 4.6.6.5. The test site shall be at ambient temperature. The following noise measurements shall be taken:

a. In a horizontal plane at the operator's position, defined to be 18 inches from the control panel, with the microphone located 1.6 meter above the floor.

b. In a horizontal plane at one meter from the perimeter of the heater at locations other than the operator's position. These measurements shall be taken following the 85 dB(A) contour specified in 4.6.6.4. Microphone shall be located 1.6 meter above the floor.

c. Readings shall be taken at 18 inches from each duct outlet. Microphone shall be located directly in line with the center of each outlet.

4.6.6.7 Information to be recorded. The following information shall be recorded:

a. Complete list of calibrated instrumentation used for measurements including: nomenclature, model, serial number, manufacturer, date of calibration, and period of calibration.

b. Complete identification of the air conditioner including: nomenclature, type, and serial number.

c. Complete description of the operational conditions under which the test was conducted.

d. Noise level measurements for background and operational tests.

4.6.6.8 Failure criteria. Nonconformance to 3.4.6 shall constitute failure of this test.

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4.6.7 Multifuel capability. The combustion burner shall be run on each of the fuels specified in 3.6.1 for not less than 8 hours. Verify that the heater operates continuously without flooding when the fuel container is 12 inches below the base of the heater and up to 10 feet above the heater. Testing shall be done within the temperature design limits of each fuel. Failure to start and operate continuously for the specified period shall constitute failure of this test.

4.6.8 Electromagnetic interference. The heater shall be subjected to the radiated emissions, electric field, 10 kHz to 18 GHz test (RE102) in accordance with MIL-STD-461 using the test plan specified in MIL-STD-462, and as specified herein.

4.6.8.1 Test conditions.

4.6.8.1.1 Steady-state. The controls of the heater shall be set for the following modes of operation at steady-state conditions:

a. "VENT" mode.
b. "AUTO HEAT" mode, highest temperature setting.
c. "AUTO HEAT" mode, lowest temperature setting.
d. "MANUAL HEAT" mode.

4.6.8.1.2 Transient. Transient durations shall be measured when the heater controls are switched between the following modes:

a. "OFF" to "VENT".
b. "VENT" to "AUTO HEAT", lowest temperature setting.
c. "AUTO HEAT" mode, temperature selection changed from lowest to highest.
d. "AUTO HEAT" mode, temperature selection changed from highest to lowest.
e. "AUTO HEAT" to "MANUAL HEAT" mode.

4.6.8.1.3 Item orientation. The radiated emissions test shall be performed in its entirety as specified herein with each of the four sides of the heater facing the antenna.

4.6.8.1.4 Failure criteria. Nonconformance to 3.5.1 shall constitute failure of this test.

4.6.9 Inclined operation. The heater shall be tilted at an angle of 10 degrees from horizontal, started and operated for 30 minutes with controls set for maximum output. This test shall be repeated for four 10-degree tilts at 90 degree increments, radially. Nonconformance with 3.7.6 shall constitute failure of this test.

4.6.10 External power.

4.6.10.1 Test procedure.

Step 1. Install the necessary instrumentation to measure volts and amperes to the heater electric power supply.

Step 2. Adjust and stabilize the environmental chamber to -40 °F (+0, -10 °F).

Step 3. Operate the heater in the Vent mode and record the inrush current. Record the voltage while the heater is operating.

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Step 4. When the heater has obtained steady state operation, measure the voltage and current. Turn the heater off after the data has been collected.

Step 5. Repeat steps 3 and 4 operating the heater in the Manual and Auto heat mode.

Step 6. Repeat steps 2 through 5 at a temperature of +65, ±3 °F.

4.6.10.2 Failure criteria. The following shall constitute failure of this test:

a. Nonconformance to 3.4.2 or 3.5.
b. Failure of the heater to start and operate.

4.6.11 Heated air quality.

4.6.11.1 Procedure. The heater shall be connected to ducts as specified in 3.10. The heater shall be operated in an ambient temperature between -40 and +65 °F. Air entering the heater shall not contain more than 15 PPM (parts per million) of carbon monoxide. The heater shall be operated for not less than 30 minutes in the heating mode. The burner must operate throughout the test.

4.6.11.2 Failure criteria. Nonconformance to 3.8.7 shall constitute failure of this test.

4.6.12 Controls. Place the unit in a properly ventilated area and visually check that all components are securely fastened in place, all connections are tight, and that the heater may be safely operated. Perform all pre-operational checks in accordance with operating instructions. After correction of any problem area discovered above, adjust the controls so that the heater runs continuously in heat mode. After not less than 5 minutes of operation, perform the following:

a. Check and adjust (as required) fuel rate for proper operation.
b. Check for evidence of fuel leaks.
c. Verify that the fuel pump is operating and fuel is delivered to the burner.
d. Check that the ignition is on and operating properly.
e. Check that the combustion air blower, circulation air fan, and other components are on and operating properly.
f. Conduct Bachrach smoke test.

Turn switch to "off" and observe the heater for proper shutdown operation. After correction of any problem areas, turn the heater on again and check the above items again. If no additional problems are discovered, operate the unit for approximately 15 minutes before conducting the following tests.

4.6.12.1 Remote thermostat control. Subject the room thermostat to the following test. Set the temperature indicator at the minimum temperature setting. Connect the thermostat to the heater and record the ambient temperature in the vicinity of the thermostat. Place the mode selector switch to the "Auto Heat" position. Increase the indicator setting until the burner ignites. Record the indicator setting temperature. After 5 minutes of operating, decrease the indicator setting until the burner shuts off. Again, record the indicator setting and ambient temperatures. Malfunctioning of components or nonconformance to 3.9.5 shall constitute failure of this test.

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4.6.12.2 Flame-failure safety. Disconnect the burner ignition system so that no combustion will occur when heater is operated in the heat mode. Start the heater with the ignition system disconnected and record the time it takes to stop the flow of fuel to the combustion chamber. After the control stops the fuel flow, reconnect the ignition system and attempt to restart the heater by depressing the manual reset control. Nonconformance to 3.8.4 or 3.9.3 shall constitute failure of this test.

4.6.12.3 High temperature safety. The heater discharge airflow shall be partially blocked during burner operation until the high temperature safety control stops the flow of fuel to the combustion chamber. After the air temperature reaches 180 °F the rate of blocking shall be adjusted to limit the rate of average temperature change to 2 °F per minute. The temperature of the discharge air shall be recorded when the safety control deactivates the fuel supply. Nonconformance to 3.8.5 or 3.9.4 shall constitute failure of this test.

4.6.13 Rail impact test.

4.6.13.1 Test conditions. This test is conducted by mounting the heater on a rail car in its rail shipment configuration. Two samples shall be used, mounted in opposite orientations. The heater shall be secured to the rail car using the four tiedown provisions of the heater. A series of at least four impacts shall be performed. The first three impacts shall be at 6.4, 9.7, and 13 km/h (4, 6, and 8 mph), respectively, in the same direction. The fourth shall be conducted at 13 km/h in the reverse direction. All four impacts shall have a tolerance of +0.8, −0.0 km/h.

4.6.13.2 Test facility. The following equipment shall be necessary to perform the rail impact test.

a. Two ordinary railroad cars, equipped with a draft gear coupling that shall be used during shipment.
b. A prime mover for the cars.

4.6.13.3 Controls. The following are minimum test control requirements.

a. A calibrated means to determine that the speed at the time of impact is 6.4, 9.7, and 13 km/hr (4, 6, and 8 mph) within ±5 percent.
b. Accelerometers and associated circuitry to measure the impact, shock, and equipment response.

4.6.13.4 Preparation for test. Preparation for test shall be as follows:

a. A visual inspection of the heater with special attention to stress areas shall be performed.
b. A operational test as specified in 4.6.2 shall be performed on the heater.
c. Results of the inspection and the operational test shall be recorded.
d. When a problem is discovered, it shall be corrected and steps a. through c. shall be repeated.

4.6.13.5 Procedure. The following procedure shall be used:

Step 1. Position from one to five rail cars on a level section of track. Satisfy the following conditions:

a. Adjust the total weight of the car assemblage to at least 114,000 kilograms (250,000 pounds).

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b. Compress the couplers between cars to take up any slack.
c. Set all of the air and hand brakes on the car(s) to be used.
d. Secure any load in or on the car(s) to prevent sliding or shifting; any movement greater than 5 cm shall be justification for retest.
e. The end of the buffer car to be struck must have a standard draft gear.

Step 2. Mount the heater on the test car. The test car shall be equipped with standard draft gears and conventional underframes. The materiel developer is responsible for the development of transportation instructions, and shall coordinate these with, and obtain approval from the Military Traffic Management Command Transportation Engineering Agency (MTMCTEA), well in advance of rail impact testing. Mounting of the test Item shall incorporate the standard loading and bracing method as shown in Section 6 of the Association of American Railroads (AAR) "Rules Governing the Loading of Department of Defense Materiel on Open Top Cars." No exotic or unusual tiedown methods shall be used; any non-standard loading and bracing must be approved by the Military Traffic Management Command Transportation Engineering Agency (MTMCTEA), ATTN: MTTE-TR, 720 Thimble Shoals Blvd. - Suite 130, Newport News, VA 23606-2574 (or its European equivalent) prior to testing.

Step 3. Situate the test car between the buffer cars and the locomotive, and pull the test car at least 65 meters (200 feet) from the buffer car(s) along a level section of track (a minimum distance to achieve the required locomotive speeds).

Step 4. Position the knuckles of the buffer and test cars for coupling.

Step 5. Install a timing device to measure the test car speed (±.1 km/h) just prior to impact with the buffer car(s). Suggested methods include electronic timing (microswitches) and radar.

Step 6. Push the test car toward the buffer car(s) and, by using the locomotive's speedometer or other means, release the test car when the desired test speed is reached, thus allowing the test car to freely impact the buffer car(s).

NOTE: Any impacts below the required test speeds shall be repeated. Impacts above the required test speed shall be accepted providing the conditions of 4.6.13.1 are satisfied. Re-testing shall be accomplished with new tiedown material to eliminate additive effects.

Step 7. Repeat Step 6 until the test car(s) has impacted the buffer car three times at the same end, once each at speeds of 6.4, 9.7, and 13.0 km/h (4, 6, and 8 mph) +0.8, -0.0 km/h. Reverse the test car and repeat the 13.0 km/hr impact for a total of 4 impacts.

NOTE: Adjustment of the lading or securing mechanisms, or reconditioning of the bracing or items of securement are not allowed during the test. If the tiedowns or chock blocks become loose during the test, a decision to completely retest shall be made.

Step 8. Repeat steps 1-7 for any other shipping orientation.

Step 9. Record the pertinent information for each impact, to include the following:

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a. Complete heater identification.
b. The number and speed of impact.
c. Observations of tiedowns, blocking, fittings, etc.
d. Observations of the test item's physical condition.

Step 10. Operate the heater in the heat mode for one hour and conduct a visual inspection. Record the results.

4.6.13.6 Information to be recorded. The following information shall be recorded:

a. Heater identification (manufacturer, serial number, etc.)
b. Previous test methods to which the heater has been subjected.
c. List of all test equipment, including vibration generating and analysis equipment, tiedown arrangements, filtering and fixtures.
d. Heater positioning with respect to the test car.

4.6.13.7 Failure criteria. The following shall constitute failure of this test:

a. Failure of the heater to start or operate after the test.
b. Malfunctioning of components or an electrical short during operation.
c. Any item that is attached to or included as an integral part of the heater breaks free, loosens or shows any sign of permanent deformation beyond specification tolerances.
d. Permanent deformation and inability to swivel or rotate the tiedown provisions
e. Nonconformance to 3.7.4.

4.6.14 Vibration test.

4.6.14.1 Test set-up. The test set-up shall be as follows:

a. The heater shall be mounted by auxiliary braces, straps, or other tiedown devices to the test apparatus. No blocking or tiedown of internal components other than those that are a part of the heater shall be employed. Removable panels and access doors shall be in place during all testing. The power cable shall be connected and arranged so that it hangs downward after being secured in the horizontal plane by three cable clips.

b. Accelerometers shall be mounted on the test apparatus immediately adjacent to the four corners of the base of the heater. The four accelerometers shall be connected into an electric or electronic network which shall average the accelerometers' output signals into a single indicated g level of the magnitude specified for this test.

c. With the heater mounted, there shall be no resonance inherent in the test apparatus at the frequencies and acceleration levels specified for this test. During the test, the base of the heater shall remain in the horizontal plane.

4.6.14.2 Preparation for test. Preparation for test shall be as follows:

a. A visual inspection of the heater shall be performed.
b. A operational test as specified in 4.6.2 shall be performed on the heater.
c. Results of the inspection and the operational test shall be recorded.
d. If a problem is discovered, it shall be corrected and steps a. through c. shall be repeated.

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4.6.14.3 Procedure. The following procedure shall be used:

a. The heater shall be exposed to 120 minutes of vibration in the transverse axis (left to right) at the levels and spectra specified in figure 6.
b. Rotate the heater 90 degrees and subject to 96 minutes of vibration in the longitudinal front (control panel end) to back (air inlet/outlet end) at the levels and spectra specified in figure 7.
c. After completion of the vibration test the heater shall be operated in the heat mode for one hour and examined for defects as specified in 3.7.3. Results of the inspection shall be recorded.

4.6.14.4 Information to be recorded. The following information shall be recorded:

a. Heater identification (manufacturer, serial number, etc.).
b. Previous test methods to which the heater has been subjected.
c. List of all test equipment, including vibration generating and analysis equipment, mounting arrangements, and fixtures.
d. Orientation of heater, including axes of applied vibration.
e. Location of accelerometers.
f. Applied test levels, duration and frequency ranges.

4.6.14.5 Failure criteria. The following shall constitute failure of this test:

a. Failure of the heater to start or operate after the test.
b. Malfunctioning of components or an electrical short during operation.
c. Nonconformance to 3.7.3.

4.6.15 Reliability/endurance/maintainability (REM) test. The REM test shall be conducted on 4 heaters for a total of 8,000 hours. After the successful completion of the operational test (see 4.6.2), each heater shall be subjected to the REM test as specified herein. Each heater shall accumulate a minimum of 2,000 hours.

4.6.15.1 Procedures. The REM test shall be conducted at an ambient temperature between -40 and +65 °F and the operation for each heater shall be subdivided in the modes below:

a. "MANUAL" heat mode: Each heater shall be subjected to 1,000 hours of testing in this mode. The heater shall be set up on an automatic test stand to operate the heater at rated output. Operation in this mode shall be continuous except as necessary to perform required scheduled maintenance. This scheduled maintenance includes the daily PMCS and scheduled maintenance at specific intervals as recommended in the manuals.

b. "AUTO" heat mode: Each heater shall be subjected to 1,000 hours of testing in this mode. The thermostat will be set so that the heater will cycle once every hour (45 minutes on and 15 minutes off). When the heater is in the "on" mode the burner must be operating. Operation in this mode shall be continuous except as necessary to perform required scheduled maintenance. This scheduled maintenance includes the daily PMCS and scheduled maintenance at specific intervals as recommended in the manuals.

4.6.15.2 Fuel types. One heater shall be operated using diesel fuel (conforming to A-A-52557) grade Low Sulfur No. 1-D, one using Low Sulfur No. 2-D, and two using JP-8 fuels during the REM test.

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4.6.15.3 Accept-reject criteria. Maintenance ratio shall be determined from data collected during the REM test.

Test duration: 8,000 hours.

Reliability: Accept if 13 or less chargeable failures.
Reject if 14 or more chargeable failures.

Chargeable failures are as defined in 6.3.4.3.

Maintainability: Accept if MR is 0.042 or less.
Reject if MR exceeds 0.042.

Endurance: Accept if 0 durability failures.
Reject if 1 or more durability failures.

Durability failures are as defined in 6.3.3.

Nonconformance to any of the acceptable criteria above or to 3.14 shall constitute failure of this test.

Maintenance ratio is as defined in 6.3.1. The services, adjustment and replacement (preventive and scheduled) shall be in accordance with 3.15.3 and 3.15.4.

4.6.15.4 Heated air quality. At the completion of the REM tests, conduct a heated air quality test, as specified in 4.6.11, on two of the test heaters.

4.6.15.5 Noise limit test. At the completion of the REM tests, conduct a noise limit test, as specified in 4.6.6, on two of the test heaters.

5. PACKAGING

5.1 Packaging. When packaging of material is to be performed by the Government, the responsible packaging activity must be contacted for the appropriate packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging department within the appropriate Military Department or Defense Agency or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity (see 5.1 & 6.2).

6. NOTES.
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The heater is intended to provide adequate temperatures in mobile shelters and similar enclosures used for both, personnel and equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).

c. When a first article is not required (see 3.2 and 4.2).

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d. When the Government will conduct any or all of the first article examinations and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests will be conducted by the contractor (see 3.2).

e. When duct assemblies are to be furnished with a built-in storage device (see 3.10.1.1).

f. When top coat color other than as specified is required (see 3.11.2).

g. When camouflage pattern painting is required (see 3.11.2).

h. When camouflage pattern data is to be provided (see 3.11.2.3).

i. When Government furnished camouflage paint pattern will be provided (see 3.11.2.4).

6.3 Definitions.

6.3.1 Maintenance ratio. The ratio of the total maintenance man-hours required (scheduled and unscheduled) to the total operating time.

6.3.2 Failure. A failure is defined as the inability of the heater to perform within the specified limits. A failure is further defined as any malfunction which:

a. Prevents the heater from starting/stopping and providing the heating output as specified in 3.3.1.

b. Causes damage to the heater or any subcomponent.

c. Causes personnel safety hazards.

6.3.3 Durability failures. Durability failures are defined as any malfunction that precludes further operation of the heater and repair is costly enough to justify scrapping major components. Durability failures are used to determine compliance with the endurance requirement of 3.14.2.

6.3.4 Failure categories. All failures occurring during REM tests will be classified as relevant and nonrelevant. Relevant failures are further classified as chargeable or nonchargeable. The contracting officer will identify all failures and classify them into the following categories:

6.3.4.1 Relevant failure. Relevant failures are defined as:

a. Intermittent failures.

b. Unverified failures (failures which cannot be duplicated, which are still under investigation, or for which no cause could be determined).

c. Verified failures not otherwise excluded.

d. Pattern failures (multiple failure occurrence of the same part in identical or equivalent applications).

6.3.4.2 Nonrelevant failure. Nonrelevant failures are defined as:

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a. Failures caused by human error (i.e., operator) where proper procedures are documented in technical manuals or on instruction plates mounted on the heater; or use of improper lubricant.

b. Failures resulting from operating the heater beyond specified limits.

c. Secondary failures which are directly caused by nonrelevant or relevant primary failures. The secondary failures must be proven to be independent on the primary failures.

d. Installation damage.

e. Failure of the test facility or test peculiar instrumentation.

f. Normal operating adjustments (nonfailures) specified in the approved equipment operating instructions.

6.3.4.3 Chargeable failures. Chargeable failures are defined as:

a. Relevant failures.
b. Independent failures related to:

- Equipment design
- Equipment manufacturing
- Part design
- Part manufacturing
- Operating, maintenance or repair procedures that cause equipment failures.

6.3.4.4 Nonchargeable failures. Nonchargeable failures are defined as:

a. Nonrelevant failures.

b. Failures induced by Government Furnished Equipment (GFE) operating, maintenance or repair procedures (i.e., prime mover failures).

c. Failures of items having a specified life expectancy and operated beyond the specified replacement time of the item.

6.4 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.11.2.4.

6.5 Subject term (keyword) listing.

Heater
Heater, chemical and biological equipment compatibility
Heater, multifuel
Heater, space
Heater, thermostatic control

6.6 Alternative chemicals for CFC-113. Several alternative cleaning agents are commercially available. The commercial sources for these agents may be obtained from the contracting officer.

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6.7 Standard color samples. Standard color samples are available from the contracting officer. The camouflage green color required in 3.10.1, 3.11.2, and 3.16.4 is similar to Camouflage Green 383 in FED-STD-595.

Custodian:
Army - CR4

Review activities:
Army - CR
DLA - CC

Preparing activity:
Army - CR4

Project 4520-A017

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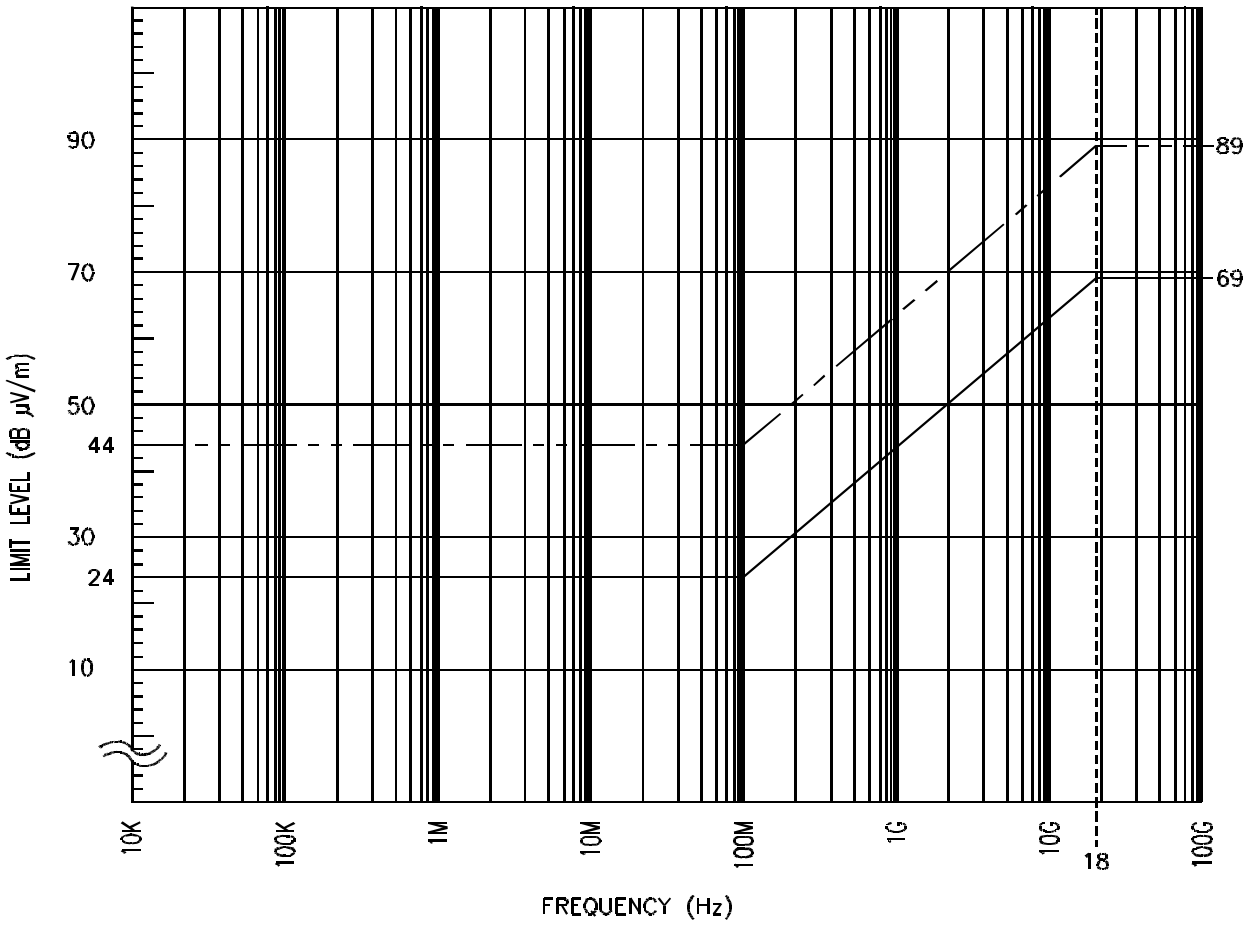
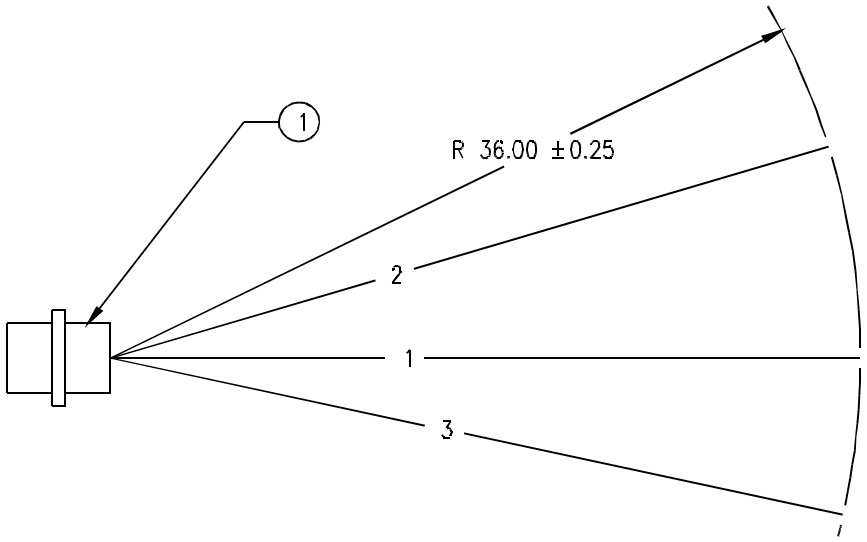


Fig. 1 RE102 LIMIT

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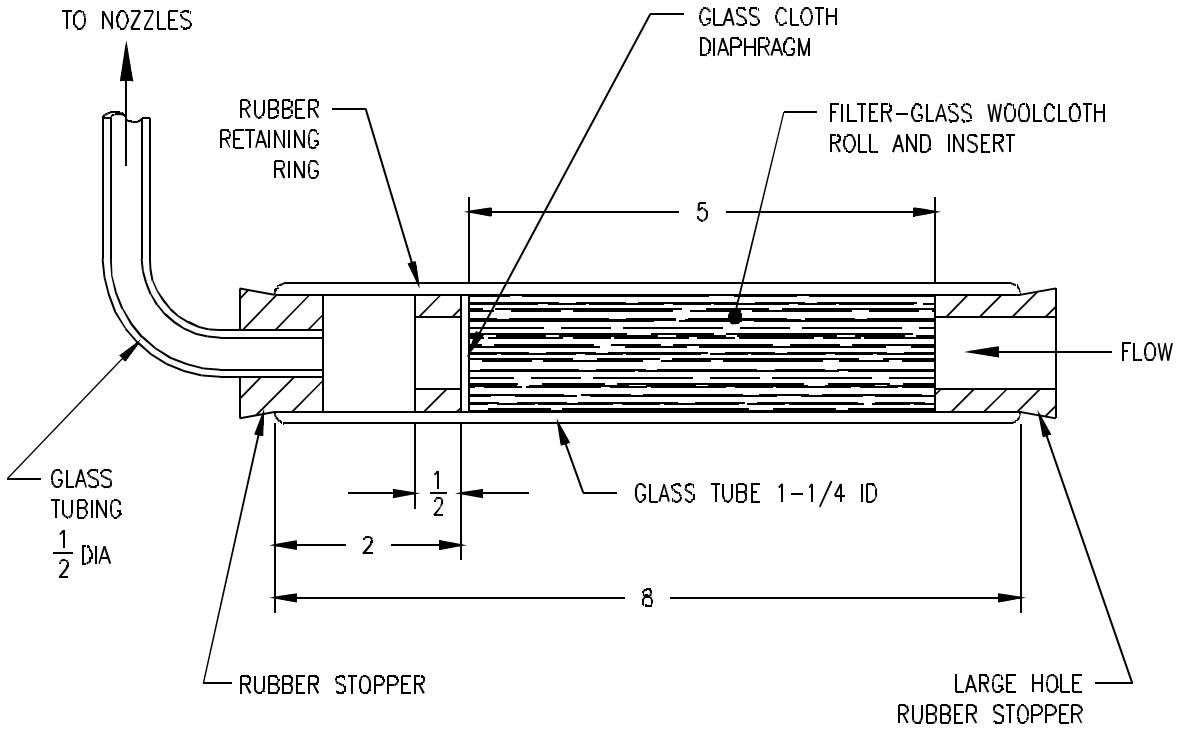
WIRE NO.	TERMINATION FIND NO.	WIRE COLOR
1	1, PIN B	WHITE
2	1, PIN A	BLACK
3	1, PIN C	GREEN

1. LEAD NOT TERMINATING IN CONNECTOR, FIND NO. 1 SHALL BE STRIPPED .50 ±.06 AND TINNED.
2. WIRE 1, 2 AND 3 SHALL BE COPPER CONDUCTOR, 600 VOLT, 105°C, WIRE SIZE 12.

1			MS3401D-16-10S	1	CONNECTORS, RECEPTACLE, ELECTRIC, FRONT		
					RELEASE, CRIMP CONTACT, AN TYPE		
FIND NO.	CAGE	DWG. SIZE	PART OR IDENTIFYING NO.	QTY REQD	NOMENCLATURE OR DESCRIPTION	SPECIFICATION	MATERIAL

Fig. 2 LEAD ASSEMBLY

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DIMENSIONS (IN INCHES) ARE FOR GUIDANCE PURPOSES.

Fig. 3 SALT SOLUTION FILTER

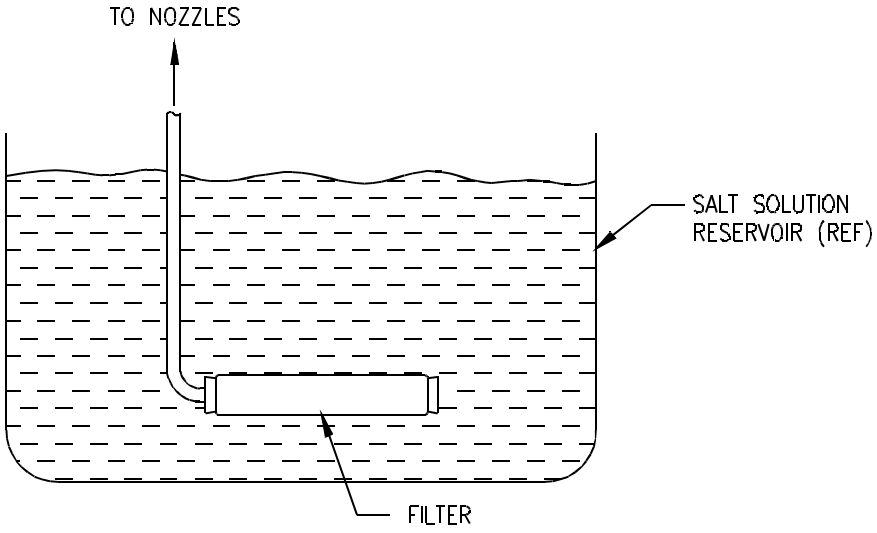


Fig. 4 LOCATION OF SALT SOLUTION FILTER

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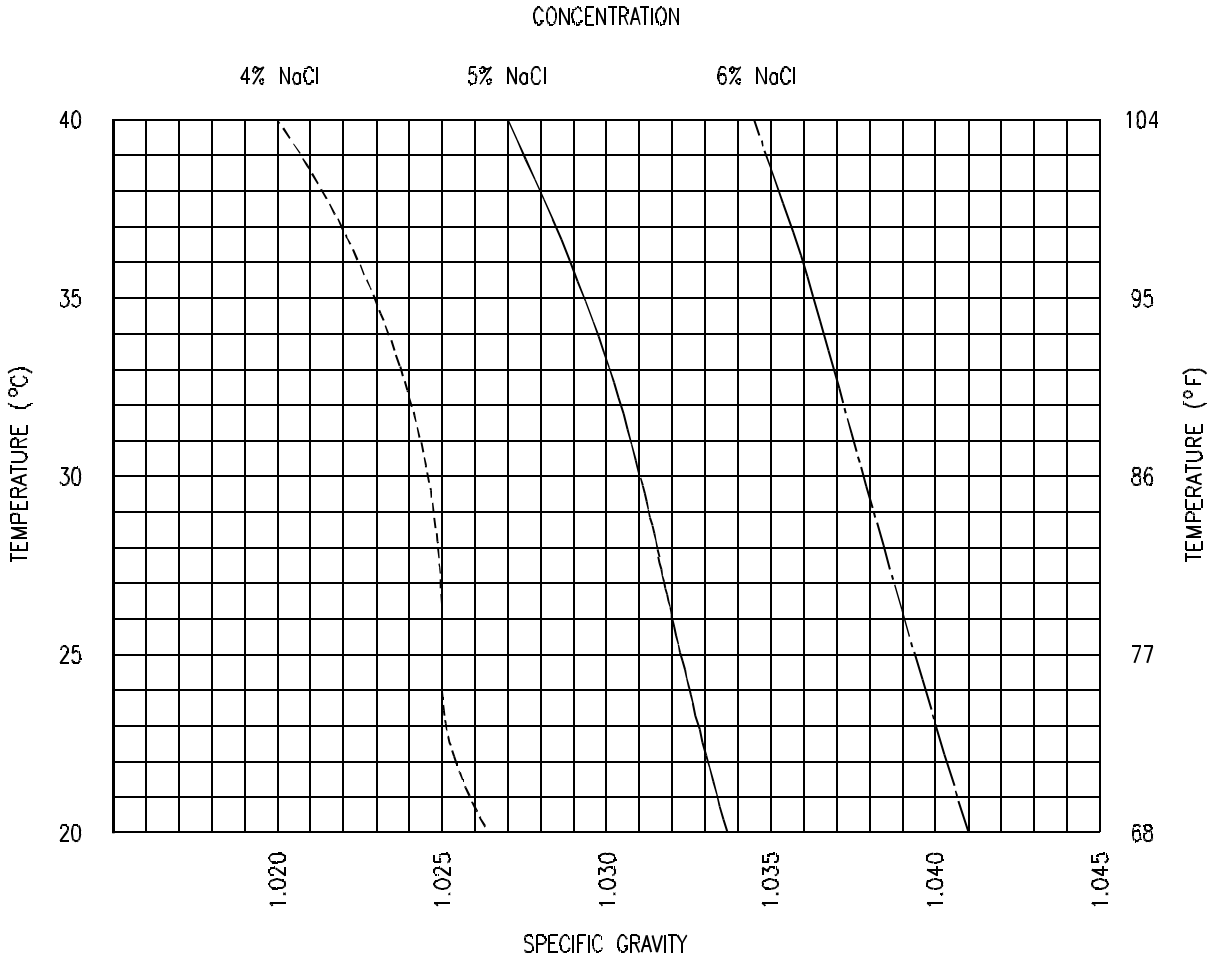


Fig. 5 VARIATIONS OF SPECIFIC GRAVITY OF SALT (NaCl) SOLUTION WITH TEMPERATURE

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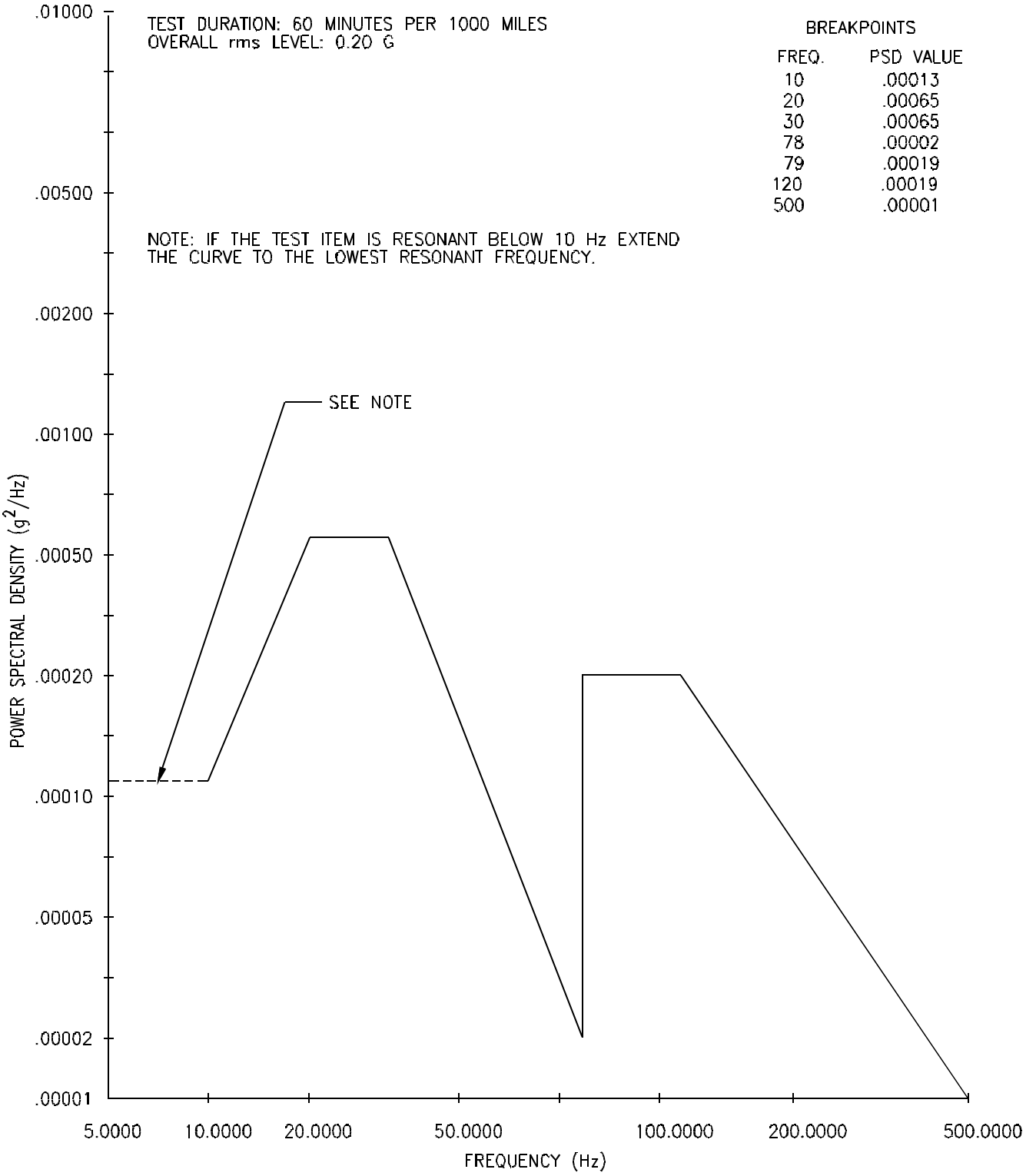


Fig. 6 BASIC TRANSPORTATION, COMMON CARRIER – ENVIROMENT. TRANSVERSE AXIS

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APPENDIX A

DUCT ASSEMBLY

A.1. SCOPE

A.1.1 Scope. This appendix covers the duct assembly to be used with the ASH.

A.2. APPLICABLE DOCUMENTS

A.2.1 Government documents.

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods.

(Unless otherwise indicated, copies of above standard is available from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D Philadelphia, PA 19111-5094.)

A.2.2 Non-Government publication. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D-471 - Rubber Property, Effects of Liquids.

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

A.3. REQUIREMENTS

A.3.1 Design. The duct assembly shall be designed for outdoor use in all types of weather to include damp, salt laden air for extended periods of time.

A.3.2 Materials. Materials selected for construction of the duct assembly shall be corrosion resistant or treated to resist corrosion caused by exposure to fuels, salt spray, or atmospheric conditions as may be encountered in storage or normal service. The protective treatment used (if any) shall in no way prevent compliance with the requirements of the specification. The use of any protective treatment that will crack, chip, peel, or scale with age or extremes of atmospheric condition shall be avoided.

A.3.3 Duct stiffener. The stiffener core shall be squared off at each end. It shall not be of such rigidity as to interfere excessively with the expansion or retraction from the storage device.

A.3.4 Cuff attachment. The stiffener core and attaching cuff shall not work loose under repeated normal usage.

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A.3.5 Dimension tolerance.

A.3.5.1 Inside diameter. The tolerances for the inside diameter of the duct assembly shall be +0.250, -0.000 inch.

A.3.5.2 Retracted outside diameter. The retracted outside diameter of the duct assembly shall not exceed 16.0 inch.

A.3.6 Environmental. The duct assembly shall meet all of the environmental requirements of the ASH with no damage or degradation of performance to the duct assembly or to the heater.

A.3.7 Fire resistance. The material used for duct assemblies shall be fire resistant to the extent that it shall not support a flame for more than 3 seconds nor glow for more than 5 seconds.

A.3.8 Airflow. The duct assembly shall be capable of delivering no less than 2,000 cubic feet per minute (cfm) of conditioned air at temperatures of up to 250 °F from the ASH to the conditioned space.

A.3.9 Friction air losses. When installed on the ASH, the air friction losses of the duct assembly shall not exceed 0.525 in wg.

A.3.10 Thermal losses. The maximum overall heat transfer coefficient (U) of the duct assembly shall not exceed 0.75 BTU/hr2-°F. Insulating material, if used, shall not wad or powder during the life of the duct, and shall not restrict the performance of the duct.

A.3.11 Expansion and retraction. Each duct assembly shall be so constructed that it will retain its approximate cross-sectional area when either expanded or retracted into the storage device. Each duct assembly shall be able to be expanded or retracted with a minimum amount of physical effort.

A.3.12 Tensile load. The duct assemblies shall be capable of withstanding a tensile pull of 250 pounds without evidence of rupture, separation of duct from connectors, or other damage.

A.3.13 Abrasion resistance. The duct assembly material shall be highly resistant to abrasion, or so constructed that the basic fabric will be protected by means of a scuff strip or the like to prevent excessive abrasion.

A.3.13 Liquid absorption. The exterior duct fabric shall be a strong, non-permeable material that has properties resistant to the absorption of gasoline, oils, grease, and water.

A.3.14 Weight. The weight of each duct assembly shall not exceed 50 pounds.

A.3.15 Product identification. In addition to that required by this document and AIA/NAS 1369, the identification marking on the duct assembly shall include the number on this specification and the contract number.

A.4 VERIFICATION

A.4.1 First article testing. In addition to the tests specified in AIA/NAS 1369, one duct assembly shall be subjected to the tests specified in A.4.2. Where there is a conflict between the two documents, the provisions of this appendix shall apply.

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A.4.1.1 Environmental testing. Unless otherwise specified, the heater shall undergo high temperature storage, low temperature, humidity, salt fog, and rain environmental testing with the duct assembly attached. Therefore the duct assembly need not be separately subjected to the above mentioned environmental tests as called out in AIA/NAS 1369.

A.4.2 Test methods.

A.4.2.1 Air flow test. Air flow test shall be conducted on full length duct assemblies with storage racks using air flow values listed in paragraphs A.3.8 and A.3.9. During testing, the duct shall be extended to full length, but shall not be placed under tension. For air friction loss tests, the inlet end of the duct assembly shall be connected to the air flow and static pressure measuring duct with the outlet of the air duct unrestricted. Friction loss values exceeding those specified in table III shall be cause for rejection.

A.4.2.2 Pressure test. One duct assembly shall be subjected to pressure tests. Blank fittings shall be attached to the end connections and shall form airtight seals when subjected to the internal pressures specified in AIA/NAS 1369. One fitting shall have provisions for regulating the internal pressure and the other shall have provisions for pressure measurement. Each duct assembly shall be fully extended when pressurized. The end connections shall be anchored to maintain at least a 90 percent extended length, when negative pressure is applied. The duct assembly shall be subjected to pressures from 8.0 psig to 5.0 in wg without rupture or other damage. Any failure to withstand these pressures shall be cause for rejection.

A.4.2.3 Insulation shift test. One duct assembly shall be tested to ensure compliance with A.3.10. A 3-foot section of each duct size shall be mounted to an extended vertical position on a vibration stand and subjected to 10 vibration cycles with a stand deflection of 0.125 inch. Immediately following the vibration test, the duct section shall be checked for an insulation shift and shall not show more than a 0.250 inch shift per foot of duct. Insulation shifts in excess of that specified shall be cause for rejection.

A.4.2.4 Bend test. A section of duct shall be submitted to 500 each, 900 degree bends over a drum, the diameter of which is twice the outside diameter size of the duct being tested. No damage shall be noted, particularly to the scuff strip, stiffener core, or attaching cuff.

A.4.2.5 Tensile load test. One duct assembly shall be subjected to a tensile load test. In testing the duct assembly, blanks shall be secured to the end connectors and the duct subjected to a tensile load of 250 pounds. All forces shall be applied to the center of the blanks. No rupture, other damage or separation of the duct assembly from connector shall occur as a result of this test. Failure of this test shall be cause for rejection.

A.4.2.6 Fire resistance test. Samples of materials used in the fabrication of the exterior surface of the duct assembly shall be tested for fire resistance in accordance with FED-STD-191, Flame Resistance of Cloth; Vertical, Field (method 5904). Nonconformance to A.3.7 and failure of this test shall be cause for rejection.

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A.4.2.7 Abrasion resistance test. Samples of materials used in the fabrication of the exterior surface of the duct assemblies shall be tested for abrasion resistance in accordance with FED-STD-191, Abrasion Resistance of Cloth; Rotary Platform, Double-Headed (Taber) Method (method 5306). Failure of this test shall be cause for rejection.

A.4.2.8 Absorption test. All materials on the exterior surface of the duct shall be tested for resistance to gasoline, oil, and water in accordance with ASTM D 471. Sample pieces of duct fabric, scuff strip fabric, and stitching shall be soaked in 91 octane gasoline, SAE No. 10 oil, and water. At the end of 48 hours, the material shall be examined to determine that the increase in thickness, weight, and volume does not exceed 10, 15, or 20 percent respectively. All other materials shall be tested in the same manner, but for resistance to water only.

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APPENDIX B

FUNGUS TEST

B.1. SCOPE

B.1.1 Scope. This appendix covers the fungus test to be used with the ASH.

B.2. APPLICABLE DOCUMENTS

B.2.1 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

B.2.1.1 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN CHEMICAL SOCIETY (ACS)

Committee on Analytical Reagents

(Application for copies should be addressed to: American Chemical Society, American Chemical Association, 1155 16th Street, NW, Washington, DC 20036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D-471 - Rubber Property, Effects of Liquids.

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

B.3 PURPOSE

B.3.1 Purpose. The purpose of the fungus test is to assess the extent to which the test item will support fungal growth or how the fungal growth may affect performance or use of the test item.

B.4 TEST

B.4.1 Test item installation. The heater shall be placed in a test facility, uncrated, packed with accessories and with panels and doors secured. Test duration shall be 28 days. The heater shall be submitted to the five species of test fungi specified in table V.

B.4.2 Test facility.

a. The required apparatus consists of chambers, together with auxiliary instrumentation, capable of maintaining and monitoring the specific conditions of temperature and humidity.

b. The chamber and accessories shall be constructed and arranged in such a manner as to prevent condensation from dripping on the test item.

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c. The chamber shall be vented to the atmosphere to prevent the buildup of pressure.

B.4.3 Controls.

a. Relative humidity shall be determined by employing either solid-state sensors whose calibration is not affected by water condensation or by an equivalent method, such as fast-reacting wet-bulb/dry-bulb sensors. Sensors that are sensitive to condensation, such as the lithium-chloride type are not recommended.

(1) When the wet-bulb control method is used, the wet-bulb assembly shall be cleaned and a new wicked installed for each test.
(2) The air velocity across the wet bulb shall not be less than 4.6 meters per second (900 feet per minute).

b. Provisions shall be made for controlling the flow of air through the internal test chamber space so that the air velocity shall be between 0.5 and 1.7 meters per second (98 to 335 ft/min).

c. Free circulation of air around the test item shall be maintained.

d. Unless otherwise specified, the test chamber temperature and relative humidity shall be recorded continuously.

e. Readout charts shall be readable to within ±0.6 °C (±1 °F).

f. The desired humidity shall be generated by using steam or water having a resistivity of not less than 250,000 ohm centimeters at 25 °C.

(1) Steam shall not be injected directly into the test chamber working space where it may have an adverse effect on the heater and microbial activity.
(2) Rust or corrosive contaminants shall not be imposed on the test item by the test facility.

g. Unless otherwise specified:

(1) All reagents shall conform to the specifications of the Committee on Analytical Reagents of the American Chemical Society, where such specifications are available.
(2) References to water shall be understood to mean distilled water or water of equal purity.

B.4.4 Test interruption. If the test is interrupted, the fact that live organisms are involved must be considered.

a. If the interruption occurs during the first seven days of the test, the test should be restarted from the beginning with a new heater or a cleaned heater to ensure a "standard test" is conducted.

b. If the interruption occurs late in the test, examine the heater for evidence of fungal growth. If the heater is bio-susceptible, there is no need for a retest. If there is no evidence of fungal growth, follow the guidance given below.

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(1) Lowered temperature. A lowering of the test chamber temperature generally will retard fungal growth. If there is no evidence of mycological deterioration and the relative humidity has been maintained, re-establish the test conditions and continue the test from the point where the temperature fell below the prescribed tolerances.

(2) Elevated temperature. Elevated temperatures may have a drastic effect on fungal growth. A complete re-initiation of the test is required if:

(a) The temperature exceeds 40 °C (104 °F), or
(b) The temperature exceeds 31 °C (88 °F) for four hours or more, or
(c) There is evidence of deterioration of the fungal colonies on the control strips.

Otherwise, reestablish test conditions and continue the test from the point of interruption.

(3) Lowered humidity. A complete re-initiation of the test is required if:

(a) The relative humidity drops below 50 percent, or
(b) The relative humidity drops below 70 percent for four hours or more, or
(c) There is evidence of deterioration of the fungal colonies on the control strips.

Otherwise, re-establish test conditions and continue the test from the point of interruption.

c. Cleaning. Any cleaning required must be conducted as least 72 hours before re-initiation and must be in accordance with B.5.2.1. New cotton control strips shall be placed in the test chamber, and both the heater and the controls will be re-inoculated with the test fungi.

B.4.5 Preparation for test.

B.4.5.1 Test duration. The test period shall be a minimum of twenty-eight days to allow for fungal germination, breakdown of carbon molecules, and degradation of material.

B.4.5.2 Pretest standard ambient checkout. A pretest standard ambient checkout shall be performed as follows:

a. The heater shall be stabilized at standard ambient conditions (see 4.6) in the test chamber.

b. Conduct a complete visual examination of the test item with special attention to discolored areas, imperfections, or the existence of any other conditions that could be conducive to fungal growth. Document the results.

c. Plugs, covers, and access plates not used in operation, but used in servicing shall remain in place.

d. An operational test as specified in 4.6.2 shall be performed and the results recorded.

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APPENDIX B
Fungus Test. - Continued

e. If the heater does not operate properly, the problem(s) shall be resolved and this checkout shall be repeated.

B.5 PROCEDURES

B.5.1 Test preparation.

B.5.1.1 Preparation of mineral-salts solution.

a. Using clean apparatus, prepare the mineral salts solution to contain the following:

Potassium dihydrogen orthophosphate (KH_2PO_4)	0.7g
Potassium monohydrogen orthophosphate (K_2HPO_4)	0.7g
Magnesium sulphate heptahydrate ($MgSO_4 \cdot 7H_20$)	0.7g
Ammonium nitrate (NH_4N0_3)	1.0g
Sodium chloride (NaCl)	0.005g
Ferrous sulfate heptahydrate ($FeSO_4 \cdot 7H_20$)	0.002g
Zinc sulfate heptahydrate ($ZnSO_4 \cdot 7H_20$)	0.002g
Manganous sulfate monohydrate ($MnSO_4\ H_20$)	0.001g
Distilled water	1000 ml

b. Measure the pH of the mineral salts solution. If it is not between 6.0 and 6.5, discard it and prepare a proper solution.

B.5.1.2 Preparation of mixed spore suspension.

NOTE - PRECAUTIONS: Although the exact strains of fungi specified for this test are not normally considered to present a serious hazard to humans, certain people say develop allergies or other reactions. Therefore standing operating procedures (SOPs) for safety should be employed. Also, the tests should be conducted by personnel trained in microbiological techniques.

a. Using aseptic techniques, prepare the spore suspension containing at least the following test fungi:

TABLE V. Test fungi.

FUNGI	FUNGUS SOURCES IDENTIFIED No.	
	USDA[1]	ATCC[2]
Aspergillus niger	QM 386	ATCC 9642
Aspergillus flavus	QM 380	ATCC 9643
Aspergillus versicolor	QM 432	ATCC 11730
Penicilliun funiculosum	QM 474	ATCC 11797
Chaetomium globosum	QM 459	ATCC 6205

[1] US Department of Agriculture, ARS Culture Collection, 1815 North University Street, Peoria, IL 61604
[2] American Type Culture Collection, 12301 Parklawn Driven, Rockville, MD 20852

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APPENDIX B
Fungus Test. - Continued

b. Maintain pure cultures of these fungi separately on an appropriate medium such as potato dextrose agar, but culture Chaetomium alobosum on strips of filter paper overlaid on the surface of mineral salts agar.

c. Prepare mineral salts agar by dissolving 15.0g of agar in a liter of the mineral salts solution described in A.3.1.1. NOTE: Do not keep the stock cultures for more than four months at 6 ±4 °C(43 ±7 °F) after that time prepare subcultures and use them for the new stocks.

d. Verify the purity of fungus cultures before the test.

e. Incubate subcultures used for preparing new stock cultures or the spore suspension at 30 ±1.4 °C (86 ±2.5 °F) for 14 to 21 days.

f. Prepare a spore suspension of each of the five fungi by pouring into one subculture of each fungus 10 ml of an aqueous solution containing 0.05g per liter of a nontoxic wetting agent such as sodium dioctyl sulfosuccinate or sodium lauryl sulfate.

g. Use a rounded glass rod to gently scrape the surface growth from the culture of the test organisms.

h. Pour the spore charge into a 125 ml capped Erlenmeyer flask containing 45 ml of water and 50 to 75 solid glass beads, 5 mm in diameter.

i. Shake the flask vigorously to liberate the spores from the fruiting' bodies and to break the spore clumps.

j. Filter the dispersed fungal spore suspension into a flask though a 6 mm layer of glass wool contained in a glass funnel.

NOTE: This process should remove large mycelial fragments and clumps of agar.

k. Centrifuge the filtered spore suspension and discard the supernatant liquid.

l. Re-suspend the residue in 50 ml of water and centrifuge. Wash the spores obtained from each of the fungi in this manner three times.

m. Dilute the final washed residue with mineral-salts solution in such a manner that the resultant spore suspension shall contain 1,000,000 ±200,000 spores per milliliter as determined with a counting chamber.

n. Repeat this operation for each organism used in the test.

o. Perform a viability check for each organism in accordance with B.5.1.3a.

p. Blend equal volumes of the resultant spore suspension to obtain the final mixed spore suspension.

NOTE: The spore suspension may be prepared fresh. If not freshly prepared, it should be held at 6 ±4 °C (43 ±7 °F) for not more than seven days.

B.5.1.3 Control items. Two types of control tests are required. Using the procedure of B.5.1.3a, verify the viability of the spore suspension and

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its preparation. By the procedure of B.5.1.3b, verify the suitability of the chamber environment.

a. Viability of spore suspension.

(1) Before preparing the composite spore suspension, inoculate sterile potato dextrose agar plates with 0.2 to 0.3 ml of the spore suspension of each of the individual fungal specifies.
(2) Distribute the inoculum over the entire surface of the plate.
(3) Incubate the inoculated potato dextrose agar plate at 24 to 31 °C (75 to 88 °F) for 7 to 10 days.
(4) After the incubation period, check the fungal growth.

NOTE: The absence of copious growth of any of the test organisms over the entire surface in each container will invalidate the results of any tests using these spores.

b. Test chamber environment.

(1) Prepare the following solution:

(a) 10.0 g glycerol.
(b) 0.1 g potassium dihydrogen orthophosphate (KH_2pO_4).
(c) 0.1 g ammonium nitrate (NH_4N0_3).
(d) 0.025g magnesium sulfate heptahydrate ($MgSO_4 \cdot 7H_20$).
(e) 0.05g yeast extract.
(f) Distilled water to a total volume of 100 ml.
(g) 0.005g of a nontoxic wetting agent such as sodium dioctyl sulfosuccinate or sodium lauryl sulfate.
(h) HCl and NaOH to adjust the final solution pH to 5.3.

(2) Dip cotton strips conforming to NATO STANAG 4370, AECTP 300, Method 308, into the above solution. After dipping, remove the excess liquid from the strips and hang them to dry before placing them in the chamber and inoculating.

(3) Within the chamber, place the strips vertically close to and bracketing the test item so that the test strips and test item experience the same test environment. The length of the strips shall be at least the height of the test item.

(4) These strips are installed and inoculated along with the test item to ensure that proper conditions are present in the incubation chamber to promote fungal growth.

B.5.2 Test performance.

B.5.2.1 Preparation for incubation.

a. Assure that the condition of the item subjected to testing is similar to the condition as delivered by the manufacturer or customer for use, or as otherwise specified. Any cleaning of the test item shall be accomplished at least 72 hours before the beginning of the fungus test.

b. Install the test item in the chamber or cabinet on suitable fixtures or suspended from hangers.

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c. Hold the test item in the operating chamber (at 24 to 31 °C (75 to 88 °F) and 95 ±5 percent RH) for at least four hours immediately before inoculation.

d. Inoculate the test item and the cotton fabric chamber control items with the mixed fungal spore suspension by spraying the suspension on the control item and on and into the test item(s) (if not permanently or hermetically sealed) in the form of a fine mist from an atomizer or nebulizer. Personnel with appropriate knowledge of the test item should be available to aid in exposing its interior surfaces for inoculation.

NOTE: In spraying the test and control items with composite spore suspension, take care to cover all external and internal surfaces which are exposed during use or maintenance. If the surfaces are non-wetting, spray until drops begin to form on them.

e. Replace covers of the test items without tightening the fasteners (so that air can penetrate).

f. Start incubation immediately following the inoculation.

B.5.2.2 Incubation of the test item.

a. Incubate the test items for the duration specified at a relative humidity of 95 ±5 percent and an air temperature of 30 ±10 °C (86 ±2 °F).

b. After seven days, inspect the growth on the control cotton strips to verify that the environmental conditions in the chamber are suitable for growth. At this time, at least 90 percent of the part of the surface area of each test strip located at the level of the test item should be covered by fungi. If it is not, repeat the entire test with the adjustments of the chamber required to produce conditions suitable for growth. Leave the control strips in the chamber for the duration of the test.

c. If the cotton strips show satisfactory fungal growth after seven days, continue the test for the required period from the time of inoculation as specified in the test plan. If there is a decrease in fungal growth on the cotton strips at the end of the test as compared to the 7-day results, the test is invalid.

B.5.2.3 Inspection. At the end of the incubation period, inspect the test item immediately. If possible, inspect the item within the chamber. If the inspection is conducted outside of the chamber and is not completed in eight hours, return the test item to the test chamber or to a similar humid environment for a minimum of 12 hours. Except for hermetically sealed equipment, open the equipment enclosure and examine both the interior and exterior of the test item. Record the results of the inspection.

NOTE: Data shall be used for comparison with the data obtained in B.3.1.

B.5.3 Operational test. Conduct an operational test as specified in 4.6.2 after the visual inspection. Data shall be recorded for comparison with the baseline data obtained in B.5.2. Personnel with appropriate knowledge of the test item should be available to aid in exposing its interior surfaces for inspection and in making operation and use decisions.

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APPENDIX B
Fungus Test. - Continued

B.6 INFORMATION TO BE RECORDED

a. Test item identification (manufacturer, serial number, etc.).
b. Presence of evidence of fungal growth at the 7-day check and at the end of the test.
c. Location of fungi.
d. Narrative description of growth, including colors, areas covered, growth patterns, density of growth, and thickness of growth (and photographs, if necessary).
e. Test period.

f. Effect of fungi on performance or use:
    (1) As received from chamber.
    (2) After removal of fungus, if appropriate.

g. Test conditions.
h. Condition of test item at time of test.

i. Deviations from specified test conditions:
    (1) Temperature.
    (2) Humidity.
    (3) Time
    (4) Air velocity.
    (5) Other.

k. Test item history (previous tests).
l. Physiological or aesthetic considerations.
m. Types of fungi used.
n. Results of performance operation tests:
    (1) Pretest.
    (2) Post-test.

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I RECOMMEND A CHANGE	1. DOCUMENT NUMBER MIL-PRF-53137A(CR4)	2. DOCUMENT DATE (YYMMDD) 970710

3. DOCUMENT TITLE HEATER, ARMY SPACE (ASH), ELECTRIC POWERED, MULTIFUEL

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

8. PREPARING ACTIVITY

a. NAME US Army Communications-Electronics Command	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3468 (2) AUTOVON 654-3468
c. ADDRESS (Include Zip Code) ATTN: AMSEL-LC-IEW-D-ED 10115 Gridley Road, Suite 228 Fort Belvoir, VA 22060-5849	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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